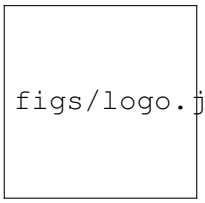
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. GATE EC 2025

A. Formulae

1.1.1 Rank of a Matrix

$$\text{rank}\,(\mathbf{A}) = \text{number of non-zero rows of } \mathbf{A} \text{ in row echelon form} \tag{1.1.1.1}$$

1.1.2 p-Series Test

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges } \iff p > 1 \tag{1.1.2.1}$$

1.1.3 Telescoping Series

$$\sum_{n=1}^{N} \frac{1}{n\,(n+1)} = \sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1} \right) \tag{1.1.3.1}$$

$$= 1 - \frac{1}{N+1} \tag{1.1.3.2}$$

1.1.4 Exponential Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \tag{1.1.4.1}$$

$$\sum_{n=1}^{\infty} \frac{1}{n!} = e - 1 \tag{1.1.4.2}$$

Solution: Substituting $x = 1$ and removing the term $n = 0$ gives (1.1.4.2).

1.1.5 Ratio Test

For a series $\sum a_n$,

$$L = \lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right| \tag{1.1.5.1}$$

$$L < 1 \implies \text{the series converges} \tag{1.1.5.2}$$

1.1.6 Expectation

$$E\,[X] = \sum_{x} x \Pr\,(X = x) \tag{1.1.6.1}$$

$$E\,[X_1 + X_2] = E\,[X_1] + E\,[X_2] \tag{1.1.6.2}$$

1.1.7 Marginal PMF

$$\Pr\,(X = x) = \sum_{y} \Pr\,(X = x, Y = y) \tag{1.1.7.1}$$

1.1.8 Independent Random Variables

If X and Y are independent,

$$\Pr\,(X = x, Y = y) = \Pr\,(X = x) \Pr\,(Y = y) \tag{1.1.8.1}$$

1.1.9 PMF of a Sum of Independent Random Variables

$$p_{X+Y}[k] = \sum_i p_X[i]\, p_Y[k-i] \tag{1.1.9.1}$$

Solution: The event $\{X+Y=k\}$ is the union over i of the disjoint events $\{X=i, Y=k-i\}$. Hence

$$\Pr(X+Y=k) = \sum_i \Pr(X=i, Y=k-i) \tag{1.1.9.2}$$

$$= \sum_i \Pr(X=i)\Pr(Y=k-i) \tag{1.1.9.3}$$

$$= \sum_i p_X[i]\, p_Y[k-i] \tag{1.1.9.4}$$

upon using (1.1.8.1).

1.10 Uncorrelated Random Variables For independent, zero-mean, unit-variance a_n and a_m, $n \neq m$,

$$E[a_n a_m] = E[a_n]\, E[a_m] = 0 \tag{1.1.10.1}$$

$$E\left[a_n^2\right] = 1 \tag{1.1.10.2}$$

$$E[a_n a_m] = \delta_{n-m} \tag{1.1.10.3}$$

1.11 Gaussian Symmetry The density of a zero-mean Gaussian random variable N is symmetric, so

$$\Pr(N > c) = \Pr(N < -c) \tag{1.1.11.1}$$

1.12 Admittances

$$Y_R = \frac{1}{R} \tag{1.1.12.1}$$

$$Y_L = \frac{1}{\mathrm{j}\omega L} \tag{1.1.12.2}$$

$$Y_C = \mathrm{j}\omega C \tag{1.1.12.3}$$

1.13 Ohm's Law for Phasors

$$\underline{I} = Y\underline{V} \tag{1.1.13.1}$$

1.14 Polar Form of Phasors

$$\mathrm{j} = 1\angle 90^\circ \tag{1.1.14.1}$$

$$-\mathrm{j} = 1\angle -90^\circ \tag{1.1.14.2}$$

$$(a\angle\theta_1)\,(b\angle\theta_2) = ab\angle(\theta_1+\theta_2) \tag{1.1.14.3}$$

1.15 Series RLC Branch

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \tag{1.1.15.1}$$

$$\omega L = \frac{1}{\omega C} \tag{1.1.15.2}$$

$$C = \frac{1}{\omega^2 L} \tag{1.1.15.3}$$

1.16 Z-Parameters of a Two-Port

$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} \tag{1.1.16.1}$$

$$z_{11} = \frac{V_1}{I_1}\Big|_{I_2=0} \tag{1.1.16.2}$$

$$z_{21} = \frac{V_2}{I_1}\Big|_{I_2=0} \tag{1.1.16.3}$$

1.17 Parallel Resistors

$$R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2} \tag{1.1.17.1}$$

1.18 LTI Systems For a system $y[n] = T\{x[n]\}$,

$$\delta[n] * h[n] = h[n] \tag{1.1.18.1}$$

$$T\{ax_1 + bx_2\} = aT\{x_1\} + bT\{x_2\} \quad \text{(linear)} \tag{1.1.18.2}$$

$$T\{0\} = 0 \quad \text{(a linear system)} \tag{1.1.18.3}$$

$$T\{x[n-k]\} = y[n-k] \quad \text{(time invariant)} \tag{1.1.18.4}$$

1.19 Fourier Transform

$$F(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-\mathrm{j}\omega t}\, \mathrm{d}t \tag{1.1.19.1}$$

1.20 Magnitude Bound of the Fourier Transform

$$|F(\omega)| \leq \int_{-\infty}^{\infty} |f(t)|\, \mathrm{d}t \tag{1.1.20.1}$$

Solution: Using $\left|\int g\right| \leq \int |g|$ and $\left|e^{-\mathrm{j}\omega t}\right| = 1$ in (1.1.19.1),

$$|F(\omega)| \leq \int_{-\infty}^{\infty} |f(t)| \left|e^{-\mathrm{j}\omega t}\right| \mathrm{d}t \tag{1.1.20.2}$$

$$= \int_{-\infty}^{\infty} |f(t)|\, \mathrm{d}t \tag{1.1.20.3}$$

1.21 Transform of $e^{-|t|}$

$$e^{-|t|} \longleftrightarrow \frac{2}{1+\omega^2} \tag{1.1.21.1}$$

Solution: From (1.1.19.1),

$$\int_{-\infty}^{\infty} e^{-|t|} e^{-\mathrm{j}\omega t}\,\mathrm{d}t = \int_{-\infty}^{0} e^{t(1-\mathrm{j}\omega)}\,\mathrm{d}t + \int_{0}^{\infty} e^{-t(1+\mathrm{j}\omega)}\,\mathrm{d}t \tag{1.1.21.2}$$

$$= \frac{1}{1-\mathrm{j}\omega} + \frac{1}{1+\mathrm{j}\omega} \tag{1.1.21.3}$$

$$= \frac{2}{1+\omega^2} \tag{1.1.21.4}$$

1.22 Fourier Series and Time Scaling A periodic signal with fundamental period T_0 has

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi k t/T_0} \tag{1.1.22.1}$$

and, for $\alpha > 0$,

$$f(\alpha t) = \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi k \alpha t/T_0} \tag{1.1.22.2}$$

$$T_y = \frac{T_0}{\alpha} \tag{1.1.22.3}$$

Solution: Replacing t by αt in (1.1.22.1) yields (1.1.22.2), with the same coefficients c_k. For $y(t) = f(\alpha t)$,

$$y\left(t + \frac{T_0}{\alpha}\right) = f(\alpha t + T_0) \tag{1.1.22.4}$$

$$= f(\alpha t) = y(t) \tag{1.1.22.5}$$

yielding (1.1.22.3).

1.23 Impulse Sampling With $\omega_s = \frac{2\pi}{T_s}$,

$$\sum_n \delta(t - \tau - nT_s) = \frac{1}{T_s} \sum_k e^{\mathrm{j}k\omega_s(t-\tau)} \tag{1.1.23.1}$$

$$Y(\omega) = \frac{1}{T_s} \sum_k e^{-\mathrm{j}k\omega_s\tau} F(\omega - k\omega_s) \tag{1.1.23.2}$$

where $y(t) = f(t) \sum_n \delta(t - \tau - nT_s)$.

Solution: Multiplying $f(t)$ by (1.1.23.1) and using that $f(t)\, e^{\mathrm{j}k\omega_s t}$ has transform $F(\omega - k\omega_s)$,

$$y(t) = \frac{1}{T_s} \sum_k e^{-\mathrm{j}k\omega_s\tau} f(t)\, e^{\mathrm{j}k\omega_s t} \tag{1.1.23.3}$$

$$Y(\omega) = \frac{1}{T_s} \sum_k e^{-\mathrm{j}k\omega_s\tau} F(\omega - k\omega_s) \tag{1.1.23.4}$$

Case (ii), $\alpha = \beta = \gamma = 0$: the last two rows are zero.

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{4.3}$$

$$\text{rank}\,(\mathbf{A}) = 2 \tag{4.4}$$

Case (iii), $\beta = \gamma = 0$ and $\alpha \neq 0$: only the fourth row is zero.

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & \alpha & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{4.5}$$

$$\text{rank}\,(\mathbf{A}) = 3 \tag{4.6}$$

Case (iv), $\alpha = \beta = \gamma \neq 0$: the matrix is already in echelon form with no zero row.

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & \alpha & \alpha \\ 0 & 0 & 0 & \alpha \end{pmatrix} \tag{4.7}$$

$$\text{rank}\,(\mathbf{A}) = 4 \tag{4.8}$$

Consider the following series:

$$\text{(i)} \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}, \qquad \text{(ii)} \sum_{n=1}^{\infty} \frac{1}{n(n+1)}, \qquad \text{(iii)} \sum_{n=1}^{\infty} \frac{1}{n!} \tag{4.9}$$

Choose the correct option. (EC 2025)

(A) Only (ii) converges
(B) Only (ii) and (iii) converge
(C) Only (iii) converges
(D) All three converge

Solution: Series (i) is a p-series, series (ii) telescopes and series (iii) passes the ratio test:

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=1}^{\infty} n^{-1/2}, \qquad p = \frac{1}{2} \leq 1 \;\Rightarrow\; \text{diverges} \qquad \text{[from (1.1.2.1)]} \tag{4.1}$$

$$\sum_{n=1}^{N} \frac{1}{n(n+1)} = 1 - \frac{1}{N+1} \;\to\; 1 \qquad \text{[from (1.1.3.2)]} \tag{4.2}$$

$$\frac{a_{n+1}}{a_n} = \frac{n!}{(n+1)!} = \frac{1}{n+1} \;\to\; 0 < 1 \qquad \text{[from (1.1.5.2)]} \tag{4.3}$$

$$\sum_{n=1}^{\infty} \frac{1}{n!} = e - 1 \qquad \text{[from (1.1.4.2)]} \tag{4.4}$$

figs/Series_convergence.png

Fig. 4.1: Partial sums of the three series

A pot contains two red balls and two blue balls. Two balls are drawn from this pot randomly without replacement. What is the probability that the two balls drawn have different colours? (EC 2025)

(A) $2/3$
(B) $1/3$
(C) $1/2$
(D) 1

Solution: Number the balls $1, 2$ (red) and $3, 4$ (blue). Choosing two balls at random is the same as selecting each ball independently with probability $\frac{1}{2}$ and keeping the outcomes with exactly two balls selected. Let B be the selection indicator of one ball, with

$p_B[0] = p_B[1] = \frac{1}{2}$. The number X of selected red balls is the sum of two independent indicators, so by the convolution formula (1.1.9.1):

$$p_X[0] = p_B[0]\,p_B[0] = \frac{1}{4} \tag{4.1}$$

$$p_X[1] = p_B[0]\,p_B[1] + p_B[1]\,p_B[0] = \frac{1}{2} \tag{4.2}$$

$$p_X[2] = p_B[1]\,p_B[1] = \frac{1}{4} \tag{4.3}$$

The number Y of selected blue balls has the same PMF, $p_Y = p_X$. The total number selected is $S = X + Y$, and (1.1.9.1) gives every case:

$$p_S[0] = p_X[0]\,p_Y[0] = \frac{1}{16} \tag{4.4}$$

$$p_S[1] = p_X[0]\,p_Y[1] + p_X[1]\,p_Y[0] = \frac{1}{4} \tag{4.5}$$

$$p_S[2] = p_X[0]\,p_Y[2] + p_X[1]\,p_Y[1] + p_X[2]\,p_Y[0] = \frac{3}{8} \tag{4.6}$$

$$p_S[3] = p_X[1]\,p_Y[2] + p_X[2]\,p_Y[1] = \frac{1}{4} \tag{4.7}$$

$$p_S[4] = p_X[2]\,p_Y[2] = \frac{1}{16} \tag{4.8}$$

Exactly two balls are drawn when $S = 2$. They have different colours when $X = 1$ and $Y = 1$, whose probability is, by independence (1.1.8.1),

Consider a frequency-modulated (FM) signal

$$f(t) = A_c \cos\left(2\pi f_c t + 3\sin\left(2\pi f_1 t\right) + 4\sin\left(6\pi f_1 t\right)\right) \tag{4.9}$$

where A_c and f_c are the amplitude and frequency (in Hz) of the carrier. The frequency f_1 is in Hz and $f_c > 100 f_1$. The peak frequency deviation of the FM signal in Hz is _____. (EC 2025)

(A) $15 f_1$
(B) $12 f_1$
(C) $4 f_1$
(D) $2 f_1$

Solution: Differentiate the phase and divide by 2π:

$$\phi(t) = 2\pi f_c t + 3\sin\left(2\pi f_1 t\right) + 4\sin\left(6\pi f_1 t\right) \tag{4.1}$$

$$f_i(t) = \frac{1}{2\pi}\frac{d\phi}{dt} = f_c + 3f_1\cos\left(2\pi f_1 t\right) + 12 f_1 \cos\left(6\pi f_1 t\right) \quad \text{[from (1.1.36.2)]} \tag{4.2}$$

$$\Delta f(t) = 3f_1\cos\left(2\pi f_1 t\right) + 12 f_1 \cos\left(6\pi f_1 t\right) \tag{4.3}$$

Both cosines equal 1 at $t = 0$, so the bound is reached:

$$\Delta f_{\text{peak}} = 3f_1 + 12 f_1 = 15 f_1 \tag{4.4}$$

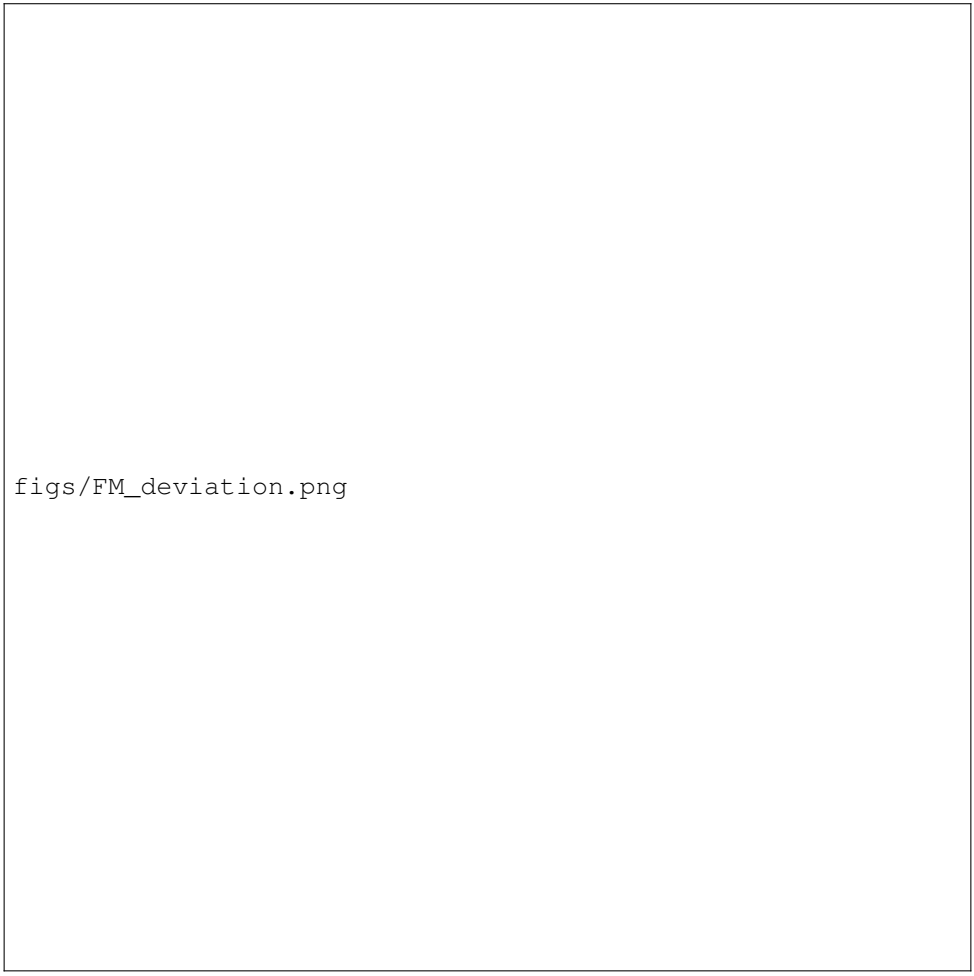


Fig. 4.1: Frequency deviation $\Delta f(t)/f_1$

Consider the discrete-time system in Fig. 4.1 with input $x[n]$ and output $y[n]$. Here $h_1[n]$ and $h_2[n]$ are the impulse responses of LTI Subsystems 1 and 2, $\delta[n]$ is the unit impulse, and $b > 0$. Assuming $h_2[n] \neq \delta[n]$, the overall system (dashed box) is _____. (EC 2025)

(A) linear and time invariant
(B) linear and time variant
(C) nonlinear and time invariant
(D) nonlinear and time variant

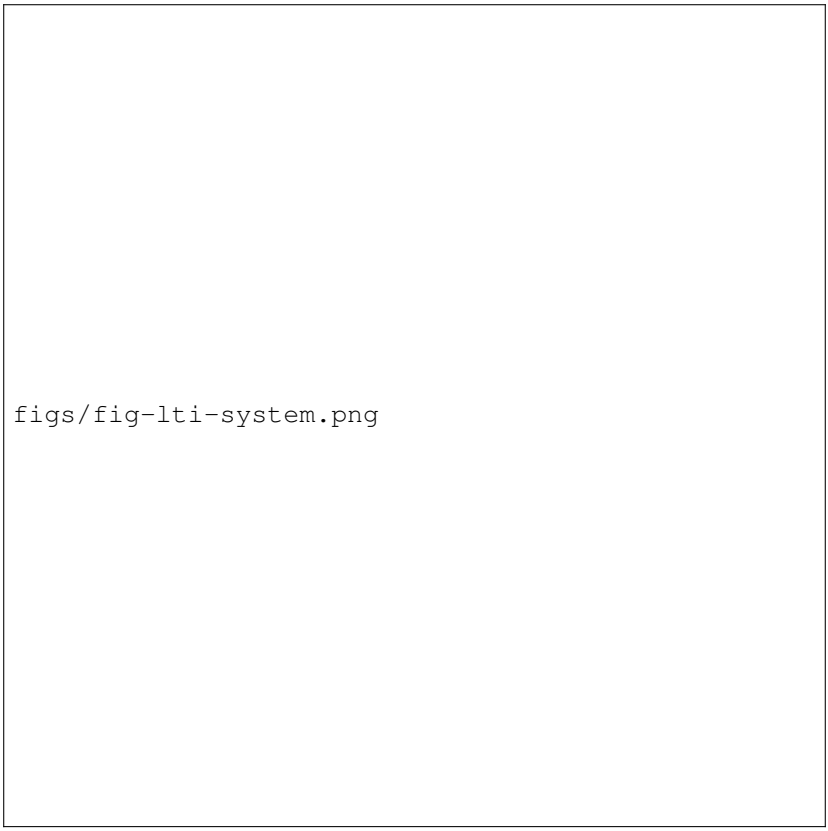


Fig. 4.1: Discrete-time system with subsystems $h_1[n]$, $h_2[n]$ and the added impulses

Solution: From the figure, $b\delta[n]$ is added after Subsystem 1 and $-b\delta[n]$ is added after Subsystem 2. Using $\delta[n] * h_2[n] = h_2[n]$ (1.1.18.1),

$$y[n] = h_2[n] * (h_1[n] * x[n] + b\delta[n]) - b\delta[n] \tag{4.1}$$

$$= (h_2 * h_1 * x)[n] + b(h_2[n] - \delta[n]) \tag{4.2}$$

For $x[n] = 0$ the output is $b(h_2[n] - \delta[n]) \neq 0$, but a linear system gives $T\{0\} = 0$ (1.1.18.3); so the system is nonlinear. For a delayed input only the first term is delayed:

$$T\{x[n-k]\} = (h_2 * h_1 * x)[n-k] + b(h_2[n] - \delta[n]) \tag{4.3}$$

$$y[n-k] = (h_2 * h_1 * x)[n-k] + b(h_2[n-k] - \delta[n-k]) \tag{4.4}$$

These differ (the term $-b\delta[n]$ stays at $n = 0$ while $\delta[n-k]$ moves, for a stable h_2), so (1.1.18.4) fails and the system is time variant.

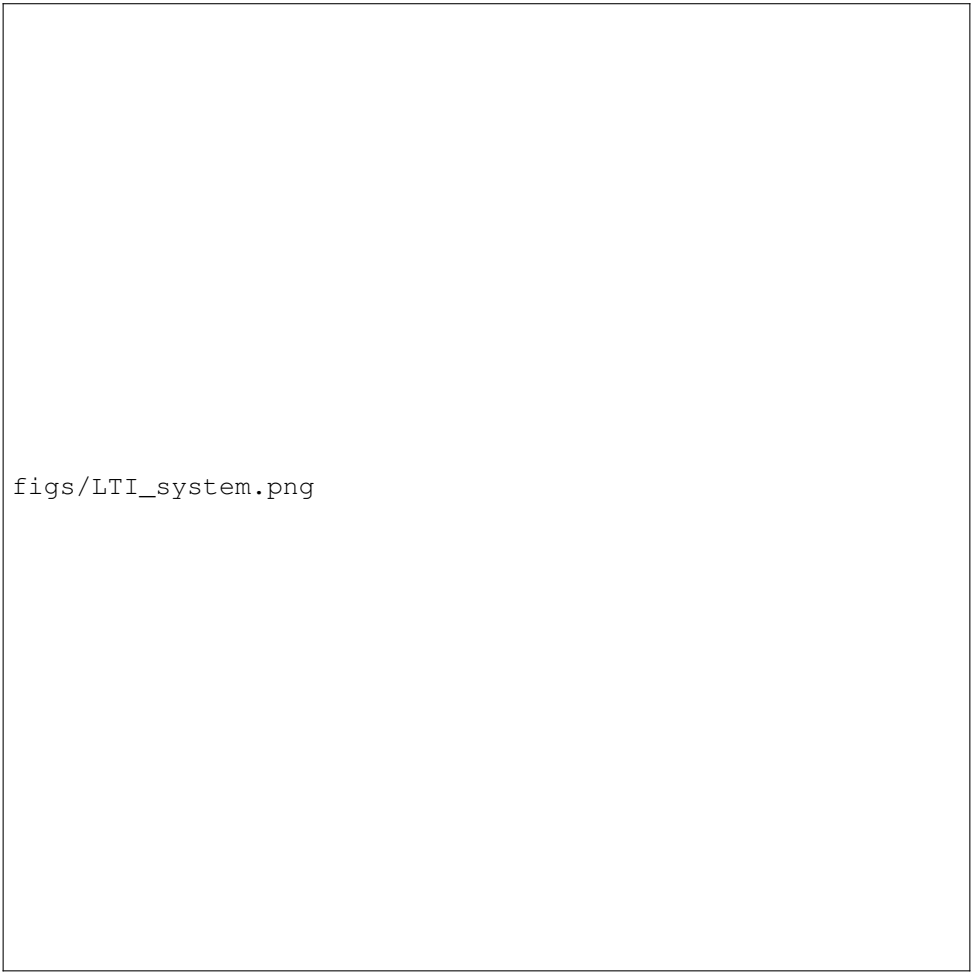


Fig. 4.2: Response to $x[n] = 0$ (left) and to a shifted input (right)

Consider a continuous-time, real-valued signal $f(t)$ whose Fourier transform $F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-\mathrm{j}\omega t}\,\mathrm{d}t$ exists. Which one of the following statements is always TRUE? (EC 2025)

(A) $|F(\omega)| \leq \int_{-\infty}^{\infty} |f(t)|\,\mathrm{d}t$

(B) $|F(\omega)| > \int_{-\infty}^{\infty} |f(t)|\,\mathrm{d}t$

(C) $|F(\omega)| \leq \int_{-\infty}^{\infty} f(t)\,\mathrm{d}t$

(D) $|F(\omega)| \geq \int_{-\infty}^{\infty} f(t)\,\mathrm{d}t$

Solution: Test the options with $f(t) = e^{-|t|}$, for which $\int |f|\,\mathrm{d}t = \int f\,\mathrm{d}t = 2$. By

(1.1.21.1),

$$|F(\omega)| = \frac{2}{1+\omega^2} \leq 2 \qquad \text{[from (1.1.21.1)]} \tag{4.1}$$

with strict inequality for $\omega \neq 0$. So (B) fails, and (D) ($|F| \geq \int f = 2$) fails. For $g(t) = -e^{-|t|}$ we get $|G(0)| = 2 > \int g\,\mathrm{d}t = -2$, so (C) fails. Option (A) holds for every signal by (1.1.20.1):

$$|F(\omega)| \leq \int_{-\infty}^{\infty} |f(t)| \left|e^{-\mathrm{j}\omega t}\right| \,\mathrm{d}t \tag{4.2}$$

$$= \int_{-\infty}^{\infty} |f(t)| \,\mathrm{d}t \qquad \text{[from (1.1.20.1)]} \tag{4.3}$$

figs/Fourier_bound.png

Fig. 4.1: $|F(\omega)|$ for $f(t) = e^{-|t|}$ against the bounds

Let i_C, i_L and i_R be the currents through the capacitor, inductor and resistor in the circuit of Fig. 4.1. The AC admittances are given in Siemens (S). Which one of the following is TRUE? (EC 2025)

(A) $i_C = 0.25\angle 180^\circ$ A, $i_L = 0.1\angle 0^\circ$ A, $i_R = 0.2\angle 90^\circ$ A

(B) $i_C = 4\angle 180^\circ$ A, $i_L = 10\angle 0^\circ$ A, $i_R = 5\angle 90^\circ$ A

(C) $i_C = 0.25\angle 270^\circ$ A, $i_L = 0.1\angle 90^\circ$ A, $i_R = 0.2\angle 90^\circ$ A

(D) $i_C = 4\angle 90^\circ$ A, $i_L = 10\angle 270^\circ$ A, $i_R = 5\angle 0^\circ$ A

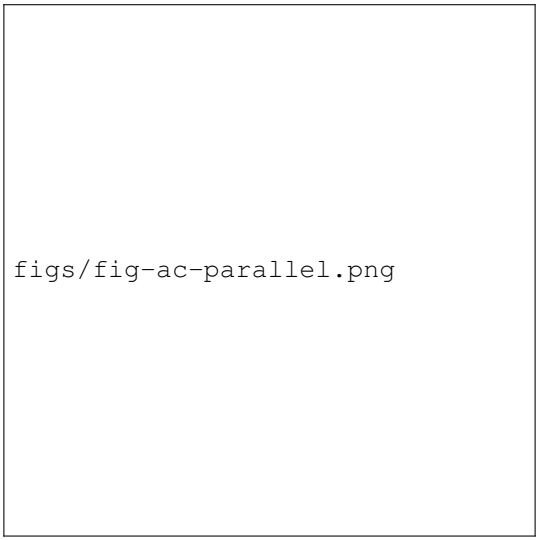


Fig. 4.1: Parallel R, L, C branches with admittances in Siemens

Solution: From the figure $\underline{V} = 1\angle 90^\circ$ V, and the branch currents are $\underline{I} = Y\underline{V}$ (1.1.13.1). Angles add when phasors are multiplied (1.1.14.3):

$$\underline{I}_C = (\text{j}0.25)\,(1\angle 90^\circ) = 0.25\angle 180^\circ \text{ A} \tag{4.1}$$
$$\underline{I}_L = (-\text{j}0.1)\,(1\angle 90^\circ) = 0.1\angle 0^\circ \text{ A} \tag{4.2}$$
$$\underline{I}_R = (0.2)\,(1\angle 90^\circ) = 0.2\angle 90^\circ \text{ A} \tag{4.3}$$

A 3-input majority logic gate has inputs X, Y, Z. The output F is logic '1' if two or more inputs are logic '1', and logic '0' if two or more inputs are logic '0'. Which one of the following is a Boolean expression of F? (EC 2025)

(A) $XY + YZ + ZX$
(B) $X \oplus Y \oplus Z$
(C) $X + Y + Z$
(D) XYZ

Solution: $F = 1$ exactly when at least two inputs are 1, i.e. for $011, 101, 110, 111$. Pairing each of the first three minterms with XYZ:

$$F = \overline{X}YZ + X\overline{Y}Z + XY\overline{Z} + XYZ \tag{4.1}$$
$$= \left(\overline{X}YZ + XYZ\right) + \left(X\overline{Y}Z + XYZ\right) + \left(XY\overline{Z} + XYZ\right) \tag{4.2}$$
$$= YZ + XZ + XY \qquad \text{[from (1.1.32.1)]} \tag{4.3}$$

The truth table (checked with bitwise operators in `Majority_gate.py`) shows that only $XY + YZ + ZX$ agrees with F in every row:

X	Y	Z	F	$XY+YZ+ZX$	$X\oplus Y\oplus Z$	$X+Y+Z$	XYZ
0	0	0	0	0	0	0	0
0	0	1	0	0	1	1	0
0	1	0	0	0	1	1	0
0	1	1	1	1	0	1	0
1	0	0	0	0	1	1	0
1	0	1	1	1	0	1	0
1	1	0	1	1	0	1	0
1	1	1	1	1	1	1	1

A full adder and an XOR gate are used to design a digital circuit with inputs X, Y, Z and output F, as in Fig. 4.1. The input Z is connected to the carry-in of the full adder. If Z is set to logic '1', the circuit functions as _____ with X and Y as inputs. (EC 2025)

(A) an adder
(B) a subtractor
(C) a multiplier
(D) a binary to Gray code converter

figs/fig-adder-xor.png

Fig. 4.1: Full adder with an XOR gate on one data input

Solution: From the figure the XOR gate forms $Y \oplus Z$ and Z is also the carry-in. For $Z = 1$ the XOR gate inverts Y, and the adder computes

$$Y \oplus Z\big|_{Z=1} = \overline{Y} \qquad \text{[from (1.1.33.2)]} \tag{4.1}$$

$$F = X + \overline{Y} + 1 \qquad \text{[from (1.1.34.1)]} \tag{4.2}$$

$$\equiv X - Y \pmod{2^n} \tag{4.3}$$

For $Z = 0$ the same circuit gives $X + Y$.

Consider $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x^3 - 3x^2 - 12x + 1$. Which of the following statements is/are correct? (EC 2025)

(A) f has no global maximizer
(B) f has no global minimizer
(C) $x = -1$ is a local minimizer of f
(D) $x = 2$ is a local maximizer of f

Solution: The critical points follow from $f'(x) = 0$, and the second derivative classifies them (1.1.48.1):

$$f'(x) = 6x^2 - 6x - 12 = 6\,(x+1)\,(x-2) \tag{4.1}$$

$$f''(x) = 12x - 6 \tag{4.2}$$

$$f''(-1) = -18 < 0 \quad \text{(local maximum, } f(-1) = 8) \tag{4.3}$$

$$f''(2) = 18 > 0 \quad \text{(local minimum, } f(2) = -19) \tag{4.4}$$

So (C) and (D) are false. Since $2x^3$ dominates,

$$\lim_{x\to\infty} f(x) = +\infty \tag{4.5}$$

$$\lim_{x\to-\infty} f(x) = -\infty \tag{4.6}$$

and f has neither a global maximizer nor a global minimizer.

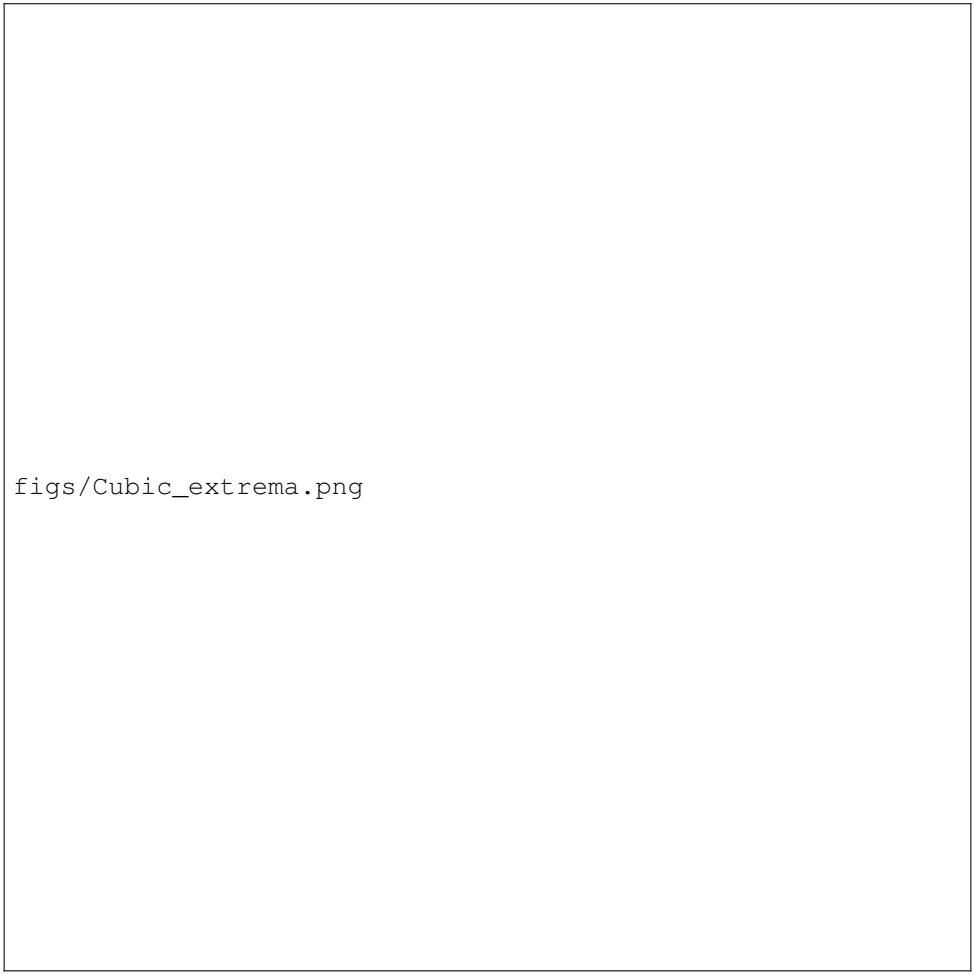


Fig. 4.1: $f(x) = 2x^3 - 3x^2 - 12x + 1$ with its local extrema

Consider the unity-negative-feedback system in Fig. 4.1, where the gain $K \geq 0$. The root locus of this system is shown in Fig. 4.2. For what value(s) of K will the system in Fig. 4.1 have a pole at $-1 + \text{j}1$? (EC 2025)

(A) $K = 5$
(B) $K = 1/5$
(C) For no positive value of K
(D) For all positive values of K

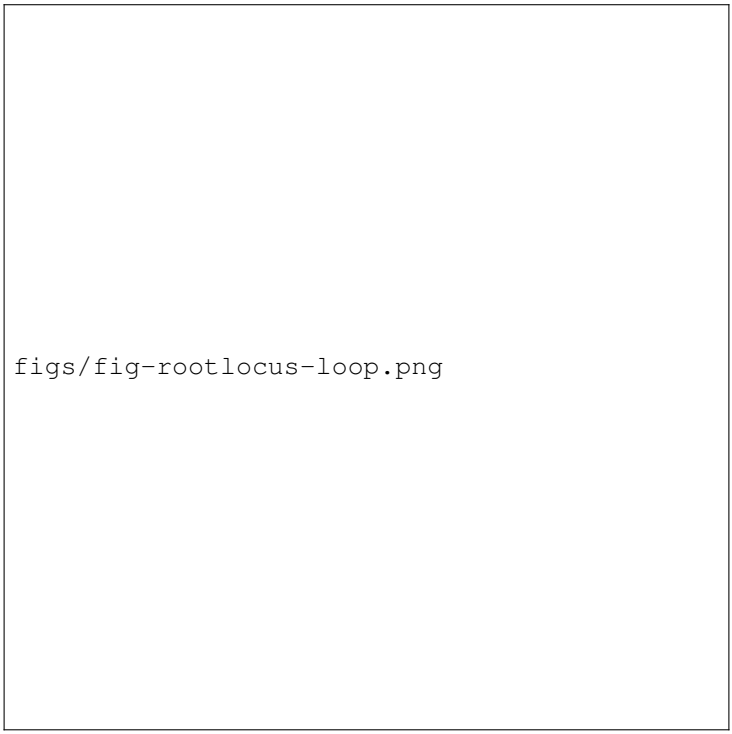


Fig. 4.1: Unity-negative-feedback system with gain K (original Figure (i))

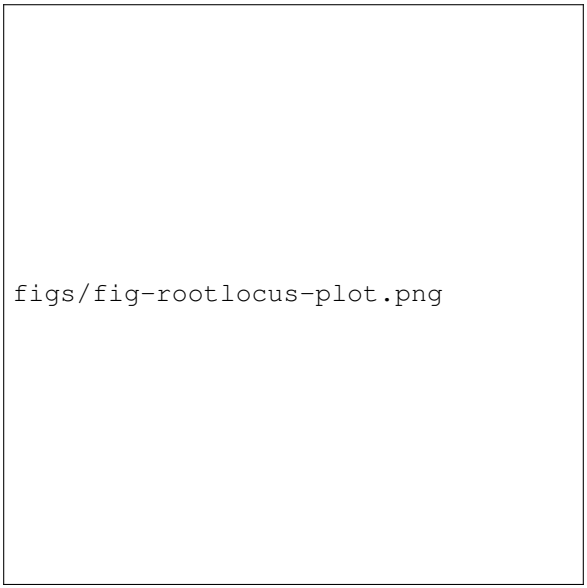


Fig. 4.2: Root locus of the system (original Figure (ii))

Solution: Fig. (ii) shows open-loop zeros at $-3, -4$ and poles at $-1, -2$, so $G(s) = \dfrac{(s+3)(s+4)}{(s+1)(s+2)}$ (a constant factor only rescales K). A point is on the locus for some $K \geq 0$ only if $G(s_0)$ is real and negative (1.1.29.1); at $s_0 = -1 + \mathrm{j}$:

$$G(s_0) = \frac{(2+\mathrm{j})(3+\mathrm{j})}{\mathrm{j}(1+\mathrm{j})} \qquad \text{[from (1.1.29.1)]} \tag{4.1}$$

$$= \frac{5+5\mathrm{j}}{-1+\mathrm{j}} \tag{4.2}$$

$$= -5\mathrm{j} \tag{4.3}$$

This is not real, so s_0 is not on the locus. The same follows from the characteristic polynomial, using $s_0^2 = -2\mathrm{j}$:

$$(1+K)\,s_0^2 + (3+7K)\,s_0 + (2+12K) = (5K-1) + \mathrm{j}\,(5K+1) \tag{4.4}$$

The real part vanishes only for $K = \frac{1}{5}$ and the imaginary part only for $K = -\frac{1}{5}$, so no K works. (The complex part of the locus is the circle $(\sigma + 2.5)^2 + \omega^2 = \frac{3}{4}$, which does not pass through s_0.)

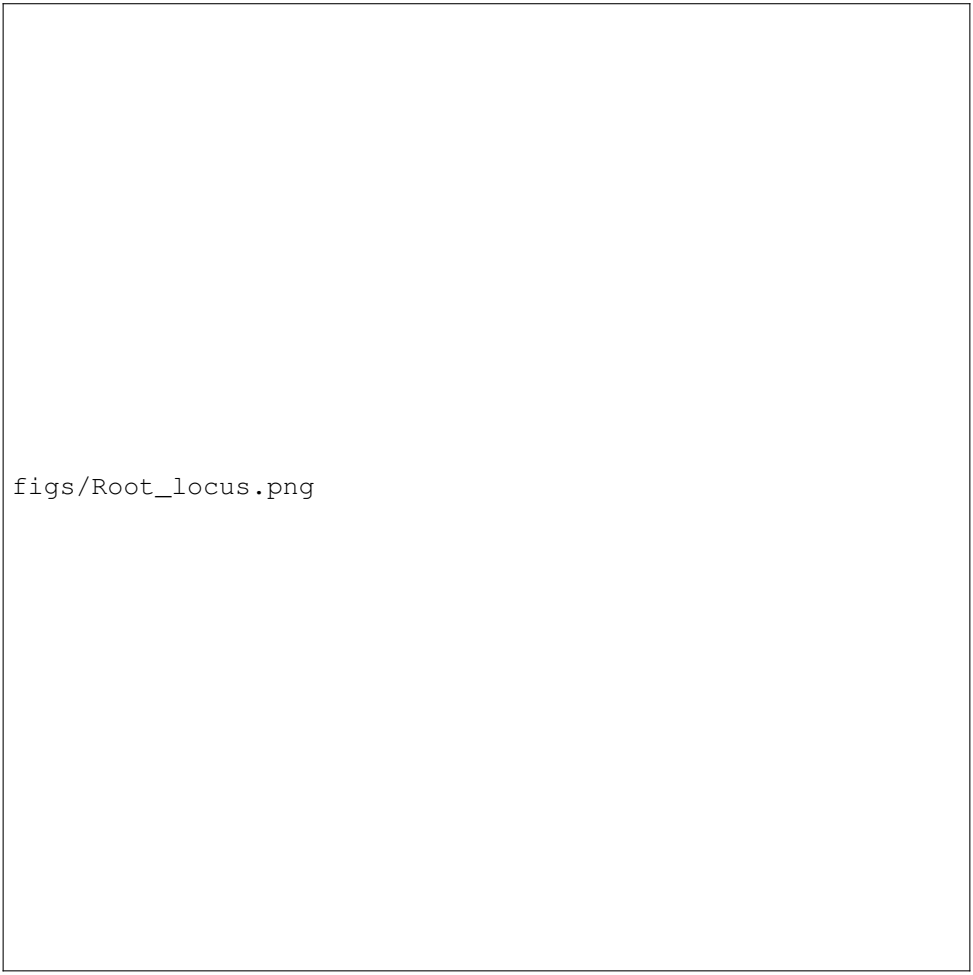


Fig. 4.3: Root locus computed from $G(s)$; the star marks $s_0 = -1 + \mathrm{j}$

Let $x[n]$ be a discrete-time signal whose z-transform is $X(z)$. Which of the following statements is/are TRUE? (EC 2025)

(A) The discrete-time Fourier transform (DTFT) of $x[n]$ always exists
(B) The region of convergence (RoC) of $X(z)$ contains neither poles nor zeros
(C) The DTFT exists if the RoC contains the unit circle
(D) If $x[n] = \alpha\delta[n]$, where $\delta[n]$ is the unit impulse and α a scalar, then the RoC is the entire z-plane

Solution: (A) is false: for $x[n] = 2^n u[n]$ the ROC is $|z| > 2$, which excludes the unit

circle, so the DTFT does not exist. (B) is false: the ROC may contain zeros, for example

$$x[n] = \delta[n] - 0.5\,\delta[n-1] \tag{4.1}$$

$$X(z) = 1 - 0.5z^{-1} = \frac{z - 0.5}{z} \tag{4.2}$$

has a zero at $z = 0.5$, inside its ROC ($z \neq 0$). (C) is true: if the ROC contains the unit circle, $X(z)$ evaluated at $z = e^{\mathrm{j}\omega}$ is the DTFT (1.1.25.2). (D) is true:

$$X(z) = \alpha \sum_n \delta[n] z^{-n} \tag{4.3}$$

$$= \alpha \qquad \text{[from (1.1.25.2)]} \tag{4.4}$$

which is finite for every z.

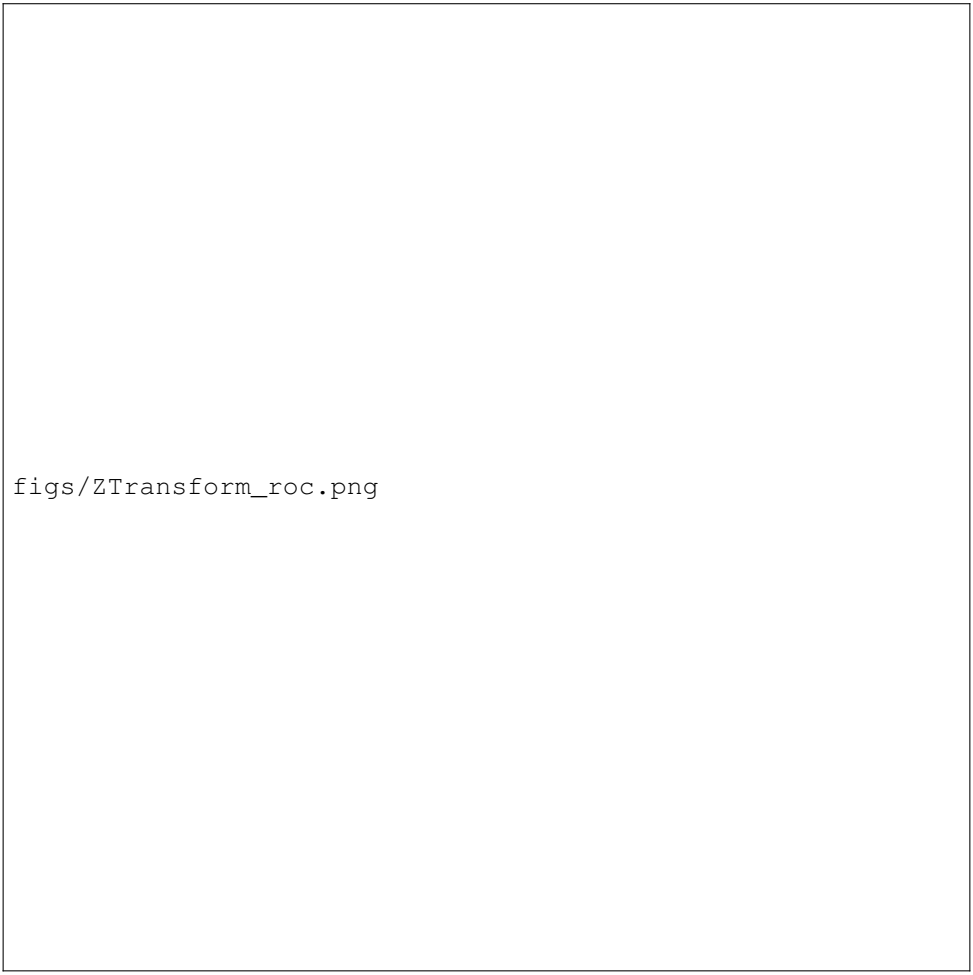


Fig. 4.1: Partial sums of $\left|x[n]e^{-\mathrm{j}\omega n}\right|$ for $2^n u[n]$ and $0.5^n u[n]$

Consider a message signal $m(t)$ bandlimited to $[-W, W]$ (W in Hz) and two schemes:

$$\text{DSB-SC:} \quad f_{DSB}(t) = A_c m(t) \cos(2\pi f_c t) \tag{4.5}$$

$$\text{AM:} \quad f_{AM}(t) = A_c \left(1 + \mu m(t)\right) \cos(2\pi f_c t) \tag{4.6}$$

Here A_c, f_c are the carrier amplitude and frequency, and μ is the modulation index. Consider the statements:

(i) An envelope detector can be used for demodulation in the DSB-SC scheme if $m(t) > 0$ for all t.

(ii) An envelope detector can be used for demodulation in the AM scheme only if $m(t) > 0$ for all t.

Which of the following options is/are correct? (EC 2025)

(A) (i) is TRUE
(B) (i) is FALSE
(C) (ii) is TRUE
(D) (ii) is FALSE

Solution: An envelope detector outputs the envelope (1.1.37.2). For DSB-SC, if $m(t) > 0$ the envelope equals the message:

$$A_c \left|m(t)\right| = A_c m(t) \qquad \text{[from (1.1.37.2)]} \tag{4.1}$$

so (i) is true. For AM the envelope is $A_c \left|1 + \mu m(t)\right|$, and recovery needs only $1+\mu m(t) \geq 0$. For $m(t) = \cos\left(2\pi W t\right)$ (negative half of the time) and $\mu = \frac{1}{2}$,

$$1 + \mu m(t) \geq \frac{1}{2} > 0 \tag{4.2}$$

so the detector works although $m(t) > 0$ fails, and (ii) is false.

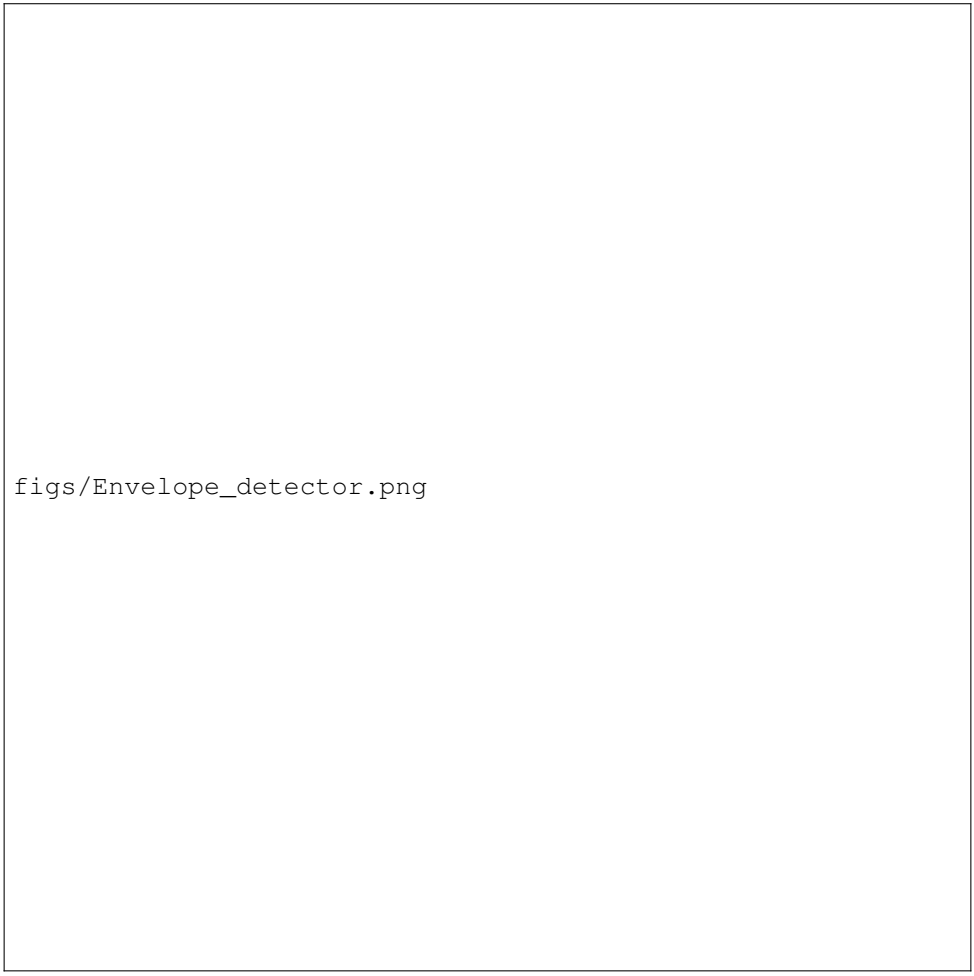


Fig. 4.1: Envelopes of the DSB-SC and AM waves for $m(t) = \cos 2\pi W t$

The function $y(t)$ satisfies $t^2y''(t) - 2ty'(t) + 2y(t) = 0$, where y' and y'' are the first and second derivatives. Given $y'(0) = 1$ and $y'(1) = -1$, the maximum value of $y(t)$ over $[0, 1]$ is _____ (rounded off to two decimal places). (EC 2025)

Solution: Try $y = t^m$ in the Euler–Cauchy equation:

$$m(m-1) - 2m + 2 = (m-1)(m-2) = 0 \qquad \text{[from (1.1.50.1)]} \tag{4.3}$$

$$y = C_1 t + C_2 t^2 \tag{4.4}$$

$$y' = C_1 + 2C_2 t \tag{4.5}$$

The conditions $y'(0) = 1$ and $y'(1) = -1$ give

$$C_1 = 1 \tag{4.6}$$

$$1 + 2C_2 = -1 \;\Rightarrow\; C_2 = -1 \tag{4.7}$$

$$y = t - t^2 \tag{4.8}$$

The maximum on $[0, 1]$ is at $y' = 0$ (1.1.48.1):

$$y' = 1 - 2t = 0 \;\Rightarrow\; t = \frac{1}{2} \tag{4.9}$$

$$y_{\max} = \frac{1}{2} - \frac{1}{4} = 0.25 \tag{4.10}$$

figs/Euler_Cauchy.png

Fig. 4.2: $y(t) = t - t^2$ and $y'(t)$

The generator matrix of a $(6, 3)$ binary linear block code is

$$\mathbf{G} = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix} \tag{4.11}$$

The minimum Hamming distance $d_{\min}$ between codewords equals _____ (answer in integer). (EC 2025)

Solution: The codewords are $\mathbf{c} = \mathbf{uG}$ (mod 2) with $\mathbf{g}_1 = 100101$, $\mathbf{g}_2 = 010011$, $\mathbf{g}_3 = 001110$:

message $\mathbf{u}$	codeword $\mathbf{c}$	weight $w(\mathbf{c})$
001	$\mathbf{g}_3 = 001110$	3
010	$\mathbf{g}_2 = 010011$	3
011	$\mathbf{g}_2 + \mathbf{g}_3 = 011101$	4
100	$\mathbf{g}_1 = 100101$	3
101	$\mathbf{g}_1 + \mathbf{g}_3 = 101011$	4
110	$\mathbf{g}_1 + \mathbf{g}_2 = 110110$	4
111	$\mathbf{g}_1 + \mathbf{g}_2 + \mathbf{g}_3 = 111000$	3

For a linear code the minimum distance is the smallest non-zero weight:

$$d_{\min} = \min_{\mathbf{c} \neq \mathbf{0}} w(\mathbf{c}) = 3 \qquad \text{[from (1.1.43.1)]} \tag{4.12}$$

All components in the bandpass filter of Fig. 4.3 are ideal. The lower -3 dB frequency is 1 MHz. The upper -3 dB frequency (in MHz, rounded off to the nearest integer) is _____. (EC 2025)



Fig. 4.3: Active bandpass filter

Solution: From the figure, the input branch is R in series with $10C$ and the feedback branch is $2R$ in parallel with $0.1C$:

$$Z_i = R + \frac{1}{s\,10C} = \frac{1 + 10sRC}{10sC} \tag{4.13}$$

$$Z_f = \frac{2R}{1 + 0.2sRC} \tag{4.14}$$

$$H(s) = -\frac{Z_f}{Z_i} = -\frac{20\,sRC}{(1 + 10sRC)\,(1 + 0.2sRC)} \qquad \text{[from (1.1.40.1)]} \tag{4.15}$$

The corner frequencies follow from the two factors (1.1.40.2), and their ratio does not depend on R, C:

$$\omega_L = \frac{1}{10RC} \qquad \text{[from (1.1.40.2)]} \tag{4.16}$$

$$\omega_H = \frac{1}{0.2RC} \qquad \text{[from (1.1.40.2)]} \tag{4.17}$$

$$\frac{f_H}{f_L} = \frac{10RC}{0.2RC} = 50 \tag{4.18}$$

$$f_H = 50 \times 1 \text{ MHz} = 50 \text{ MHz} \tag{4.19}$$

(The exact -3 dB points measured from the peak gain are 0.97 and 51.7 MHz; the question uses the corner frequencies.)

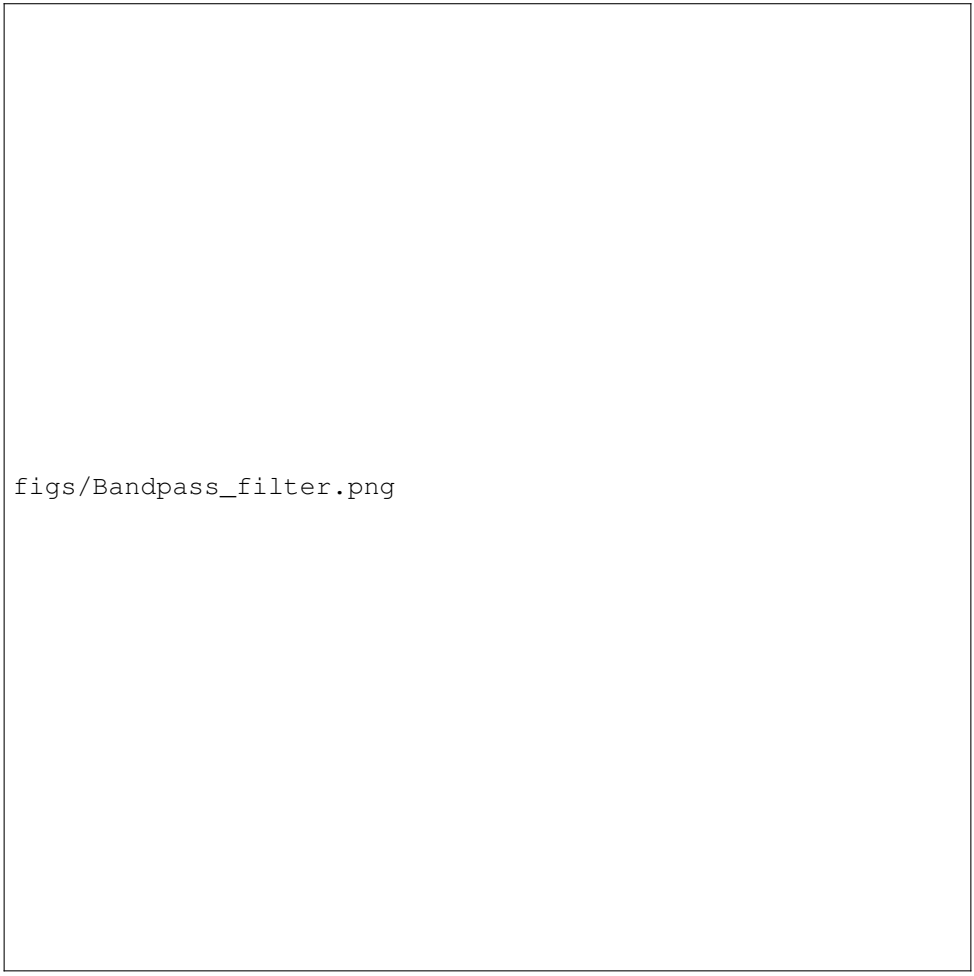


Fig. 4.4: Magnitude response of the bandpass filter

Let $G(s) = \dfrac{1}{10s^2}$ be the transfer function of a second-order system. A controller $M(s)$ is connected to $G(s)$ in the configuration of Fig. 4.1. Consider:

(i) There exists no controller $M(s) = \dfrac{K_I}{s}$, with K_I a positive real number, such that the closed-loop system is stable.

(ii) There exists at least one controller $M(s) = K_P + sK_D$, with K_P, K_D positive real numbers, such that the closed-loop system is stable.

Which one of the following options is correct? (EC 2025)

(A) (i) is TRUE and (ii) is FALSE
(B) (i) is FALSE and (ii) is TRUE
(C) Both (i) and (ii) are FALSE
(D) Both (i) and (ii) are TRUE

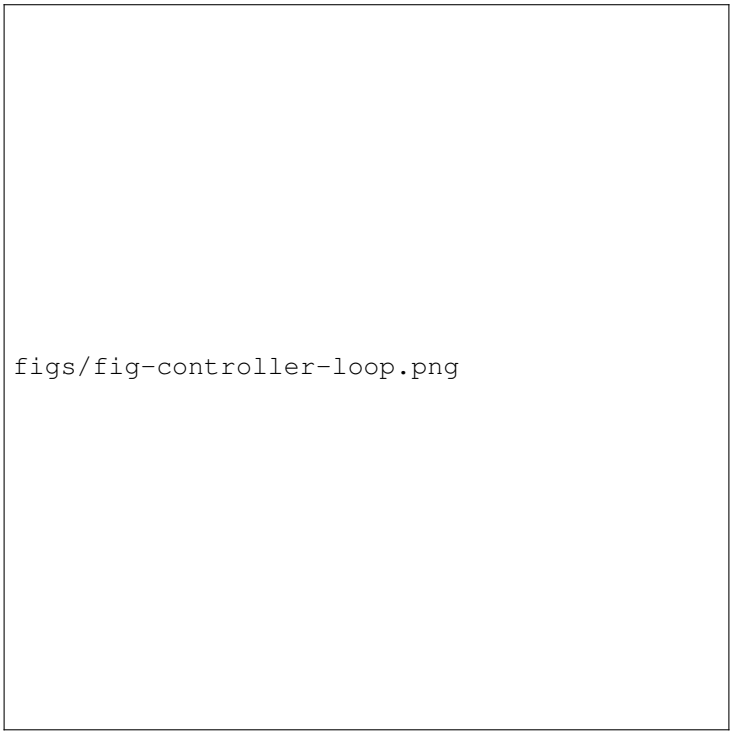


Fig. 4.1: Controller $M(s)$ and plant $G(s)$ in feedback

Solution: From the figure the loop is unity negative feedback with $M(s)$ and $G(s)$ in series, so the characteristic equation is $1 + M(s)G(s) = 0$ (1.1.26.1). For $M(s) = K_I/s$,

$$1 + \frac{K_I}{s} \cdot \frac{1}{10s^2} = 0 \qquad \text{[from (1.1.26.1)]} \tag{4.1}$$

$$10s^3 + K_I = 0 \tag{4.2}$$

$$s = \left(\frac{K_I}{10}\right)^{1/3} e^{\mathrm{j}(\pi+2\pi k)/3} \qquad \text{[from (1.1.28.1)]} \tag{4.3}$$

The roots are at angles $60°, 180°, 300°$, so two lie in the right half-plane for every $K_I > 0$: (i) is true. For $M(s) = K_P + sK_D$,

$$1 + \frac{K_P + sK_D}{10s^2} = 0 \tag{4.4}$$

$$10s^2 + K_D s + K_P = 0 \tag{4.5}$$

All coefficients are positive, so this is stable (1.1.27.1): (ii) is true.

figs/Controller_stability.png

Fig. 4.2: Closed-loop poles: integral controller (left), PD controller (right)

Consider the polynomial $p(s) = s^5 + 7s^4 + 3s^3 - 33s^2 + 2s - 40$. Let (L, I, R) be defined as: L is the number of roots of $p(s)$ with negative real parts, I is the number of purely imaginary roots, and R is the number of roots with positive real parts. Which one of the following is correct? (EC 2025)

(A) $L = 2$, $I = 2$, and $R = 1$
(B) $L = 3$, $I = 2$, and $R = 0$
(C) $L = 1$, $I = 2$, and $R = 2$
(D) $L = 0$, $I = 4$, and $R = 1$

Solution: Expand the product of the factors step by step:

$$(s+5)(s+4) = s^2 + 9s + 20 \tag{4.1}$$

$$\left(s^2 + 9s + 20\right)(s-2) = s^3 + 7s^2 + 2s - 40 \tag{4.2}$$

$$\left(s^3 + 7s^2 + 2s - 40\right)\left(s^2 + 1\right) = s^5 + 7s^4 + 3s^3 - 33s^2 + 2s - 40 \tag{4.3}$$

So $p(s) = (s+5)(s+4)(s-2)\left(s^2+1\right)$. The roots $-5, -4$ have negative real part, $\pm$j are purely imaginary and 2 has positive real part:

$$(L, I, R) = (2, 2, 1) \tag{4.4}$$

The graph of $p(s)$ crosses the real axis at the three real roots $-5, -4, 2$.

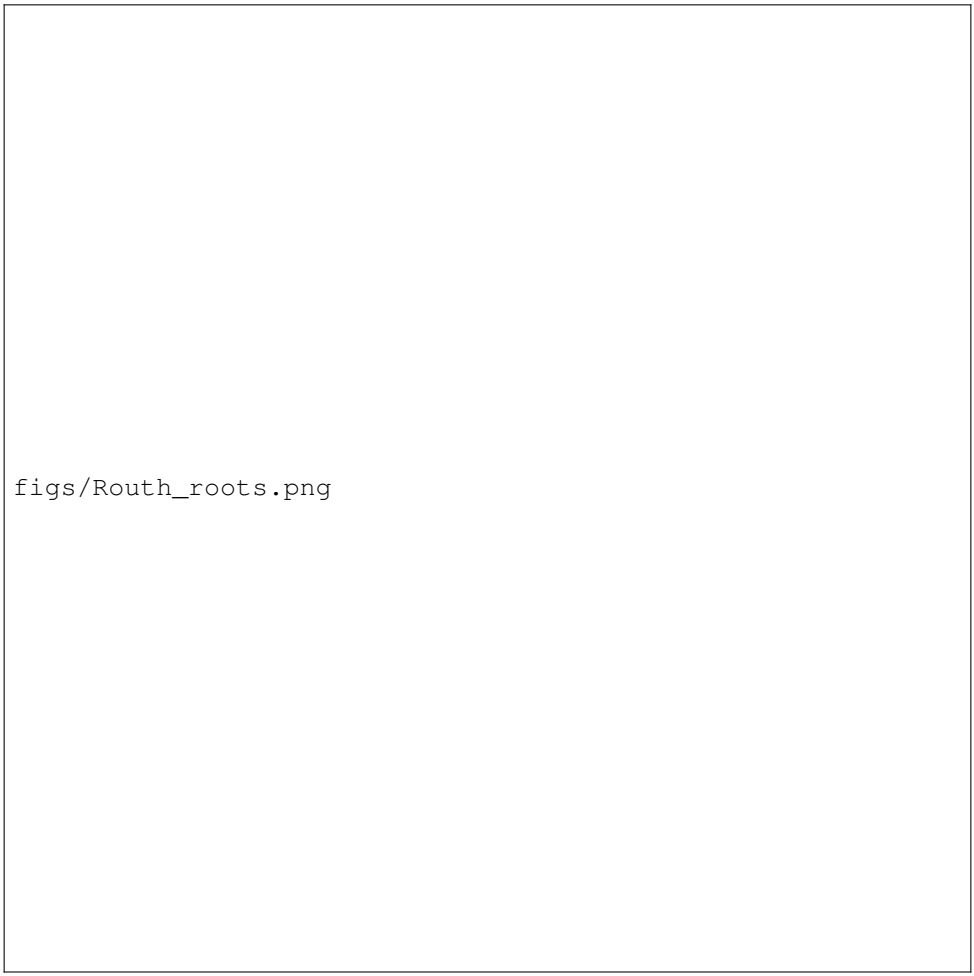


Fig. 4.1: Graph of $p(s) = s^5 + 7s^4 + 3s^3 - 33s^2 + 2s - 40$ with its real roots

Consider a continuous-time finite-energy signal $f(t)$ whose Fourier transform vanishes outside $[-\omega_c, \omega_c]$ (ω_c in rad/s). It is uniformly sampled to obtain $y(t) = f(t)p(t)$, where

$$p(t) = \sum_{n=-\infty}^{\infty} \delta\left(t - \tau - nT_s\right), \qquad T_s > 0,\ \tau > 0 \tag{4.5}$$

and $\delta(t)$ is the Dirac impulse. The sampled signal $y(t)$ is passed through an ideal lowpass filter

$$h(t) = \omega_c T_s \frac{\sin\left(\omega_c t\right)}{\pi \omega_c t} \tag{4.6}$$

with cutoff frequency ω_c and passband gain T_s. The output of the filter is _____. (EC 2025)

(A) $f(t)$ if $T_s < \pi/\omega_c$
(B) $f(t-\tau)$ if $T_s < \pi/\omega_c$
(C) $f(t-\tau)$ if $T_s < 2\pi/\omega_c$
(D) $T_s f(t)$ if $T_s < 2\pi/\omega_c$

Solution: Multiplying $f(t)$ by the shifted impulse train gives the spectrum (1.1.23.2), where the delay τ only appears as a phase on the replicas $k \neq 0$:

$$Y(\omega) = \frac{1}{T_s} \sum_{k} e^{-\mathrm{j}k\omega_s\tau} F\left(\omega - k\omega_s\right) \qquad \text{[from (1.1.23.2)]} \tag{4.1}$$

The lowpass filter (gain T_s, cutoff ω_c) keeps only $k = 0$ when the replicas do not overlap:

$$\omega_s = \frac{2\pi}{T_s} > 2\omega_c \iff T_s < \frac{\pi}{\omega_c} \qquad \text{[from (1.1.24.2)]} \tag{4.2}$$

$$H(\omega)Y(\omega) = T_s \cdot \frac{1}{T_s} F(\omega) = F(\omega) \qquad \text{[from (1.1.24.1)]} \tag{4.3}$$

so the output is $f(t)$. The samples are values of f itself on a shifted grid, so there is no delay.

figs/Sampling_offset.png

Fig. 4.1: Reconstruction equals $f(t)$, not $f(t-\tau)$

In the circuit of Fig. 4.1, M_1 is an ideal AC voltmeter and M_2 an ideal AC ammeter. The source voltage (in Volts) is $v_s(t) = 100\cos(200t)$. What should be the value of the variable capacitor C such that the RMS readings on M_1 and M_2 are 25 V and 5 A, respectively? (EC 2025)

(A) 25 μF
(B) 4 μF
(C) 0.25 μF
(D) Insufficient information to find C

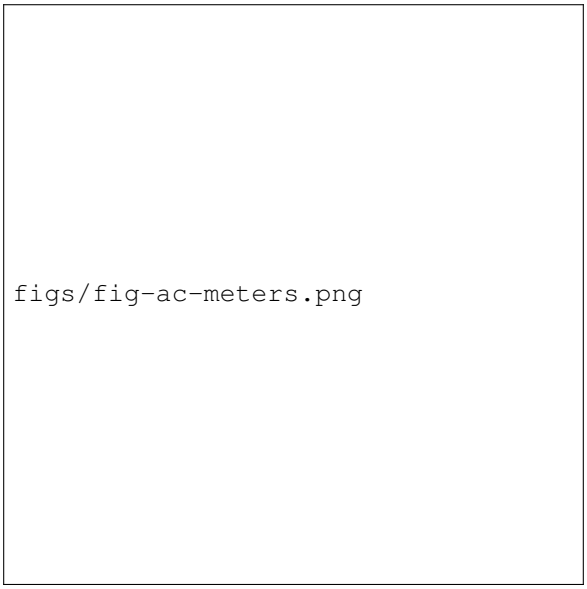


Fig. 4.1: Circuit with voltmeter M_1, ammeter M_2 and variable capacitor C

Solution: From the figure, M_1 reads the voltage across the branch 5 Ω, C, 1 H and M_2 reads its current:

$$|Z| = \frac{V}{I} = \frac{25}{5} = 5\ \Omega \tag{4.1}$$

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = R = 5\ \Omega \qquad \text{[from (1.1.15.1)]} \tag{4.2}$$

So the net reactance is zero, and with $\omega = 200$ rad/s, $L = 1$ H:

$$C = \frac{1}{\omega^2 L} = \frac{1}{200^2 \times 1} = 25\ \mu\text{F} \qquad \text{[from (1.1.15.3)]} \tag{4.3}$$

The Z-parameter matrix of a two-port network relates port voltages and currents by

$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \mathbf{Z} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} \tag{4.4}$$

The Z-parameter matrix (each entry in Ohms) of the network in Fig. 4.1 is _____. (EC 2025)

(A) $\begin{pmatrix} 10/3 & 2/3 \\ 2/3 & 10/3 \end{pmatrix}$

(B) $\begin{pmatrix} 2/3 & 10/3 \\ 10/3 & 2/3 \end{pmatrix}$

(C) $\begin{pmatrix} 10 & 2 \\ 2 & 10 \end{pmatrix}$

(D) $\begin{pmatrix} 10/3 & 1/3 \\ 1/3 & 10/3 \end{pmatrix}$

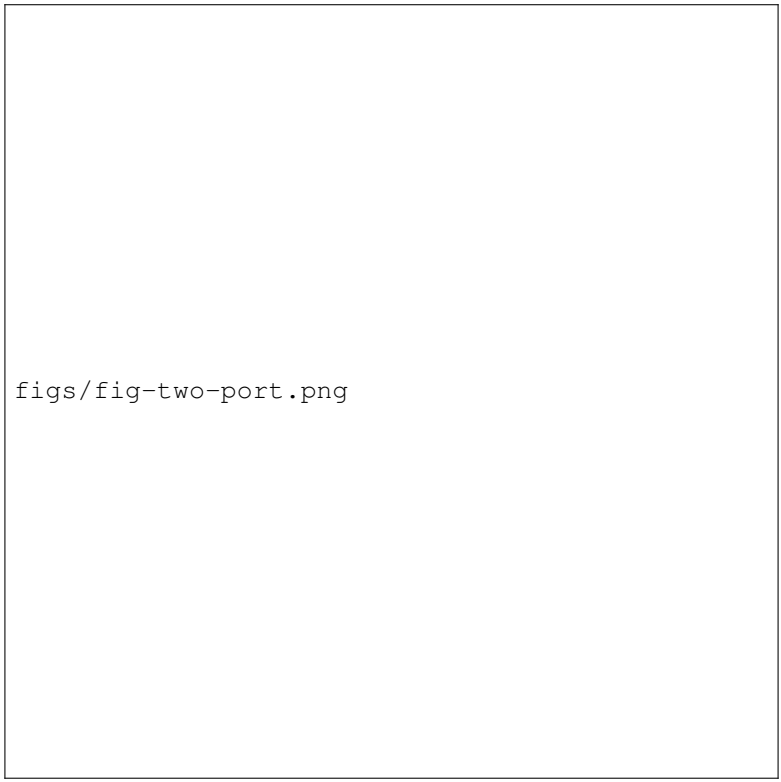


Fig. 4.1: Resistive two-port network

Solution: The figure is a ladder of 2 Ω resistors. With port 2 open ($I_2 = 0$) no current flows in the last series resistor. Seen from node A, the shunt 2 Ω is in parallel with $2+2$:

$$R_A = 2 \parallel (2+2) = \frac{2 \cdot 4}{2+4} = \frac{4}{3}\ \Omega \qquad \text{[from (1.1.17.1)]} \tag{4.1}$$

$$z_{11} = \frac{V_1}{I_1} = 2 + R_A = \frac{10}{3}\ \Omega \qquad \text{[from (1.1.16.3)]} \tag{4.2}$$

The port-2 voltage is the divider of $V_A = I_1 R_A$ through the series 2 Ω and the shunt 2 Ω:

$$V_2 = \frac{2}{2+2} V_A = \frac{1}{2} \cdot \frac{4}{3} I_1 = \frac{2}{3} I_1 \tag{4.3}$$

$$z_{21} = \frac{V_2}{I_1} = \frac{2}{3}\ \Omega \qquad \text{[from (1.1.16.3)]} \tag{4.4}$$

The network is reciprocal and symmetric, so $z_{12} = z_{21}$ and $z_{22} = z_{11}$:

$$\mathbf{Z} = \begin{pmatrix} 10/3 & 2/3 \\ 2/3 & 10/3 \end{pmatrix}\ \Omega \qquad \text{[from (1.1.16.1)]} \tag{4.5}$$

A source transmits a symbol S taking values uniformly at random from $\{-2, 0, 2\}$. The receiver obtains $Y = S + N$, where N is a zero-mean Gaussian random variable independent of S, and uses the maximum-likelihood decoder. Suppose the probability of symbol error is $P_e = \alpha \Pr(N > 1)$, where $\Pr(N > 1)$ is the probability that N exceeds 1. What is α? (EC 2025)

(A) $1/3$
(B) 1
(C) $2/3$
(D) $4/3$

Solution: ML detection picks the nearest symbol (1.1.38.1), so the decision boundaries are the midpoints -1 and 1. Let $p = \Pr(N > 1)$; by symmetry $\Pr(N < -1) = p$ (1.1.11.1). Then

$$\Pr(\text{error} \mid S = -2) = \Pr(N > 1) = p \tag{4.1}$$

$$\Pr(\text{error} \mid S = 0) = \Pr(|N| > 1) = 2p \qquad \text{[from (1.1.11.1)]} \tag{4.2}$$

$$\Pr(\text{error} \mid S = 2) = \Pr(N < -1) = p \tag{4.3}$$

$$P_e = \frac{1}{3}(p + 2p + p) = \frac{4}{3}p \tag{4.4}$$

so $\alpha = \frac{4}{3}$.

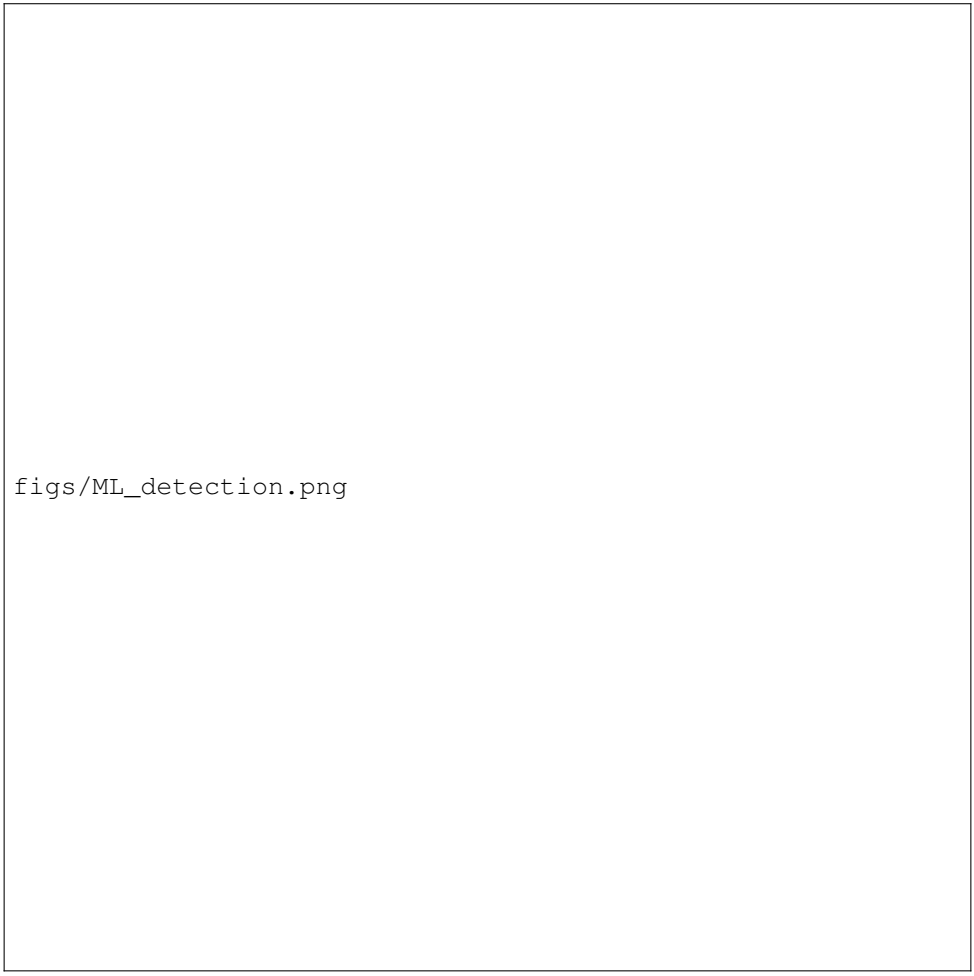


Fig. 4.1: Likelihoods of the three symbols and the ML thresholds

Consider the real-valued random process

$$f(t) = \sum_{n=1}^{N} a_n \, p\,(t - nT) \tag{4.5}$$

where $T > 0$, N is a positive integer, $p(t) = 1$ for $t \in [0, 0.5T]$ and 0 otherwise, and the coefficients a_n are pairwise independent, zero-mean, unit-variance random variables. Consider:

(i) The mean of the process $f(t)$ is independent of time t.

(ii) The autocorrelation $E[f(t)f(t+\tau)]$ is independent of t for all τ.

Choose the correct option. (EC 2025)

(A) (i) is TRUE and (ii) is FALSE
(B) Both (i) and (ii) are TRUE
(C) Both (i) and (ii) are FALSE
(D) (i) is FALSE and (ii) is TRUE

Solution: Since $E[a_n] = 0$, the mean is zero for every t, so (i) is true:

$$E\left[f(t)\right] = \sum_{n=1}^{N} E\left[a_n\right] p\left(t - nT\right) = 0 \qquad \text{[from (1.1.6.2)]} \tag{4.1}$$

The cross terms of the autocorrelation vanish because $E[a_n a_m] = \delta_{n-m}$ (1.1.10.3). For $\tau = 0$, using $p^2 = p$:

$$E\left[f(t)f(t+\tau)\right] = \sum_{n=1}^{N} p\left(t - nT\right)\, p\left(t + \tau - nT\right) \qquad \text{[from (1.1.10.3)]} \tag{4.2}$$

$$E\left[f(t)^2\right] = \sum_{n=1}^{N} p\left(t - nT\right) \in \{0, 1\} \tag{4.3}$$

It equals 1 for $t \in [nT,\, nT + 0.5T]$ and 0 just after, so it depends on t and (ii) is false.

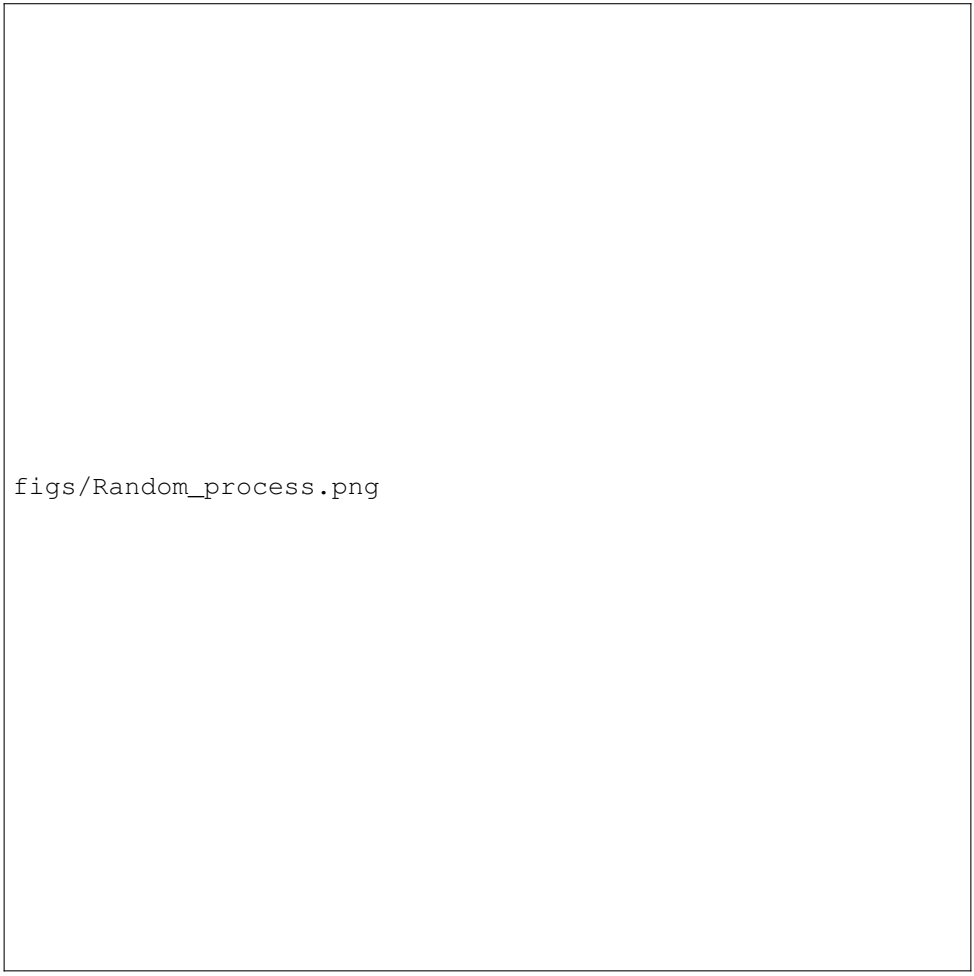


Fig. 4.1: $E[f(t)] = 0$ but $E[f(t)^2]$ depends on t

A 10-bit analog-to-digital converter (ADC) has a sampling frequency of 1 MHz and a full-scale voltage of 3.3 V. For an input sinusoid of frequency 500 kHz, the maximum SNR (in dB, two decimal places) and the data rate (in Mbps) at the ADC output are _____, respectively. (EC 2025)

(A) 61.96 and 10
(B) 61.96 and 5
(C) 33.36 and 10
(D) 33.36 and 5

Solution: For $N = 10$ bits and $f_s = 1$ MHz (1.1.39.2):

$$\text{SNR}_{\max} = 6.02\,(10) + 1.76 = 61.96 \text{ dB} \quad \text{[from (1.1.39.2)]} \tag{4.1}$$

$$R_b = f_s N = \left(10^6\right)(10) = 10 \text{ Mbps} \quad \text{[from (1.1.39.2)]} \tag{4.2}$$

figs/ADC_snr.png

Fig. 4.1: Maximum SNR of an ideal ADC against the number of bits

A positive-edge-triggered sequential circuit is shown in Fig. 4.1. There are no timing violations. Input P_0 is logic '0' and P_1 is logic '1' at all times. The timing diagram of SEL and S is also shown. The sequence of output Y from time T_0 to T_3 is _____. (EC 2025)

(A) 1011
(B) 0100
(C) 0010
(D) 1101

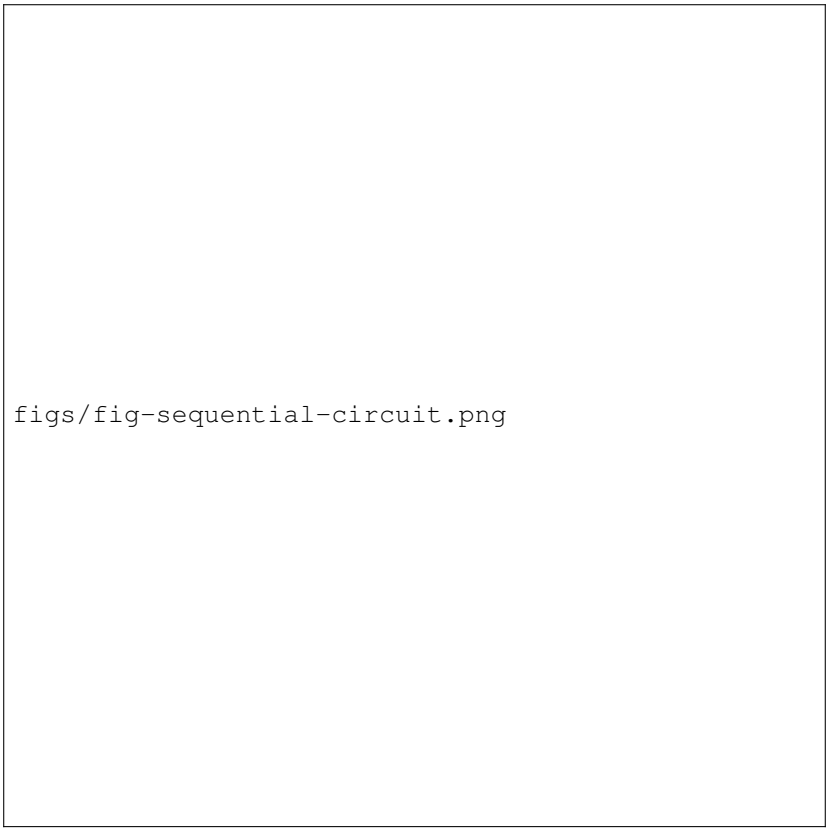


Fig. 4.1: Sequential circuit with multiplexers and flip-flops, and timing diagram of SEL and S

Solution: From the figure, with $P_0 = 0$ and $P_1 = 1$, each flip-flop copies its input at a rising edge (1.1.35.1), and the gate output is $Y = \overline{Q_1} \oplus \overline{Q_2} = Q_1 \oplus Q_2$:

$$Q_1^+ = \begin{cases} S, & SEL = 1 \\ 0, & SEL = 0 \end{cases} \quad \text{[from (1.1.35.1)]} \tag{4.1}$$

$$Q_2^+ = \begin{cases} Q_1, & SEL = 1 \\ 1, & SEL = 0 \end{cases} \quad \text{[from (1.1.35.1)]} \tag{4.2}$$

$$Y = Q_1 \oplus Q_2 \tag{4.3}$$

The marks $T_0, \ldots, T_3$ are at the falling edges of CLK, so Y is read just after the rising edge before each mark. Reading SEL and S at those four rising edges and updating (Q_2 uses the old Q_1):

edge before	SEL	S	Q_1	Q_2	Y
T_0	0	0	0	1	1
T_1	1	0	0	0	0
T_2	1	1	1	0	1
T_3	0	1	0	1	1

The first edge has $SEL = 0$, which fixes the state, so the initial state does not matter. The output sequence is $Y = 1011$.

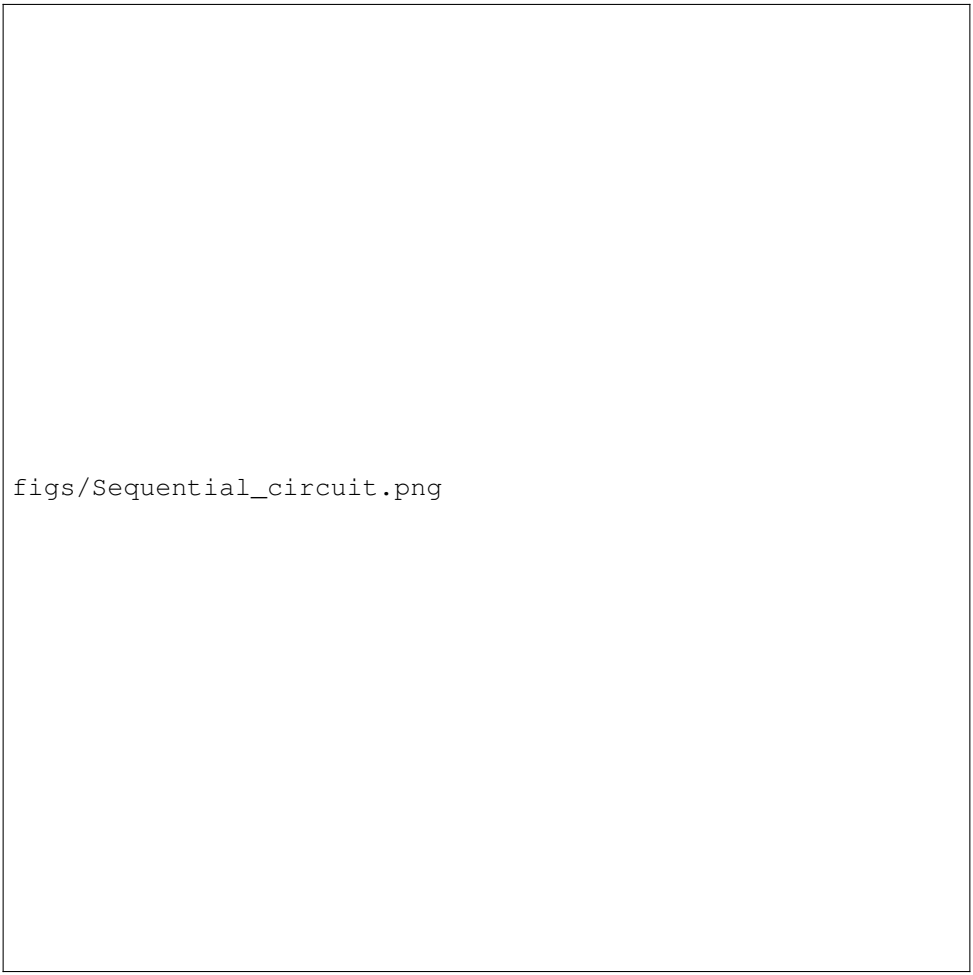


Fig. 4.2: Simulated waveforms of CLK, SEL, S, Q_1, Q_2 and Y

A square metal sheet of $4\ \text{m} \times 4\ \text{m}$ is placed on the x–y plane as in Fig. 4.1. If the surface charge density (in μC/m^2) is $\rho_s(x, y) = 4\,|y|$, the total charge (in μC, rounded off to the nearest integer) on the sheet is _____. (EC 2025)

(A) 16
(B) 85
(C) 64
(D) 256

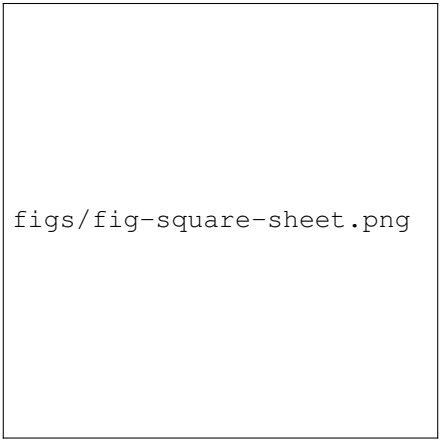


Fig. 4.1: Square metal sheet in the x–y plane

Solution: From the figure the sheet is the square $-2 \le x \le 2$, $-2 \le y \le 2$ (in m). The integrand separates and $|y|$ is even:

$$Q = \int_{-2}^{2}\int_{-2}^{2} 4\,|y|\;\mathrm{d}x\,\mathrm{d}y \qquad \text{[from (1.1.46.1)]} \tag{4.1}$$

$$= 4\left(\int_{-2}^{2}\mathrm{d}x\right)\left(2\int_{0}^{2} y\,\mathrm{d}y\right) \tag{4.2}$$

$$= 4\cdot 4\cdot 4 = 64\ \mu\mathrm{C} \tag{4.3}$$

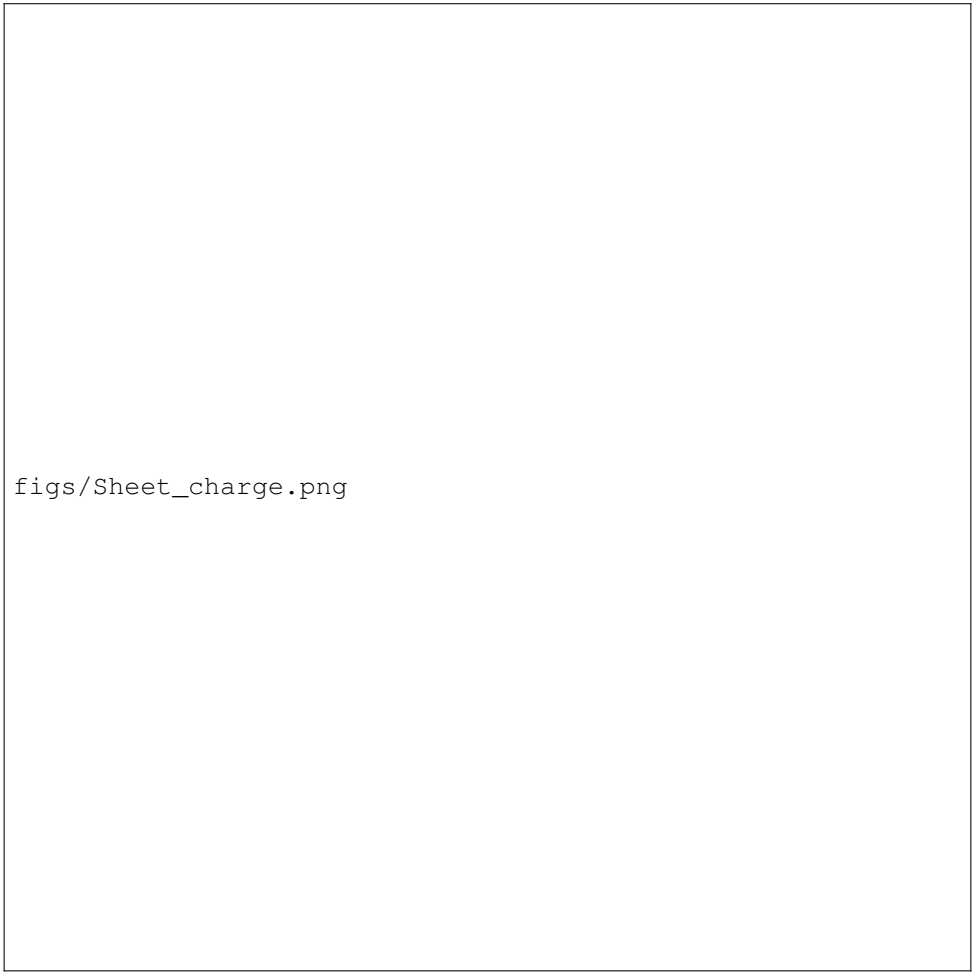


Fig. 4.2: Charge density $\rho_s = 4\,|y|$ on the sheet

An electric field of 0.01 V/m is applied along the length of a copper wire of circular cross-section with diameter 1 mm. Copper has conductivity 5.8×10^7 S/m. The current (in Amperes, two decimal places) through the wire is _____. (EC 2025)

(A) 0.46
(C) 0.58
(B) 1.82
(D) 1.12

Solution: The current density is $J = \sigma E$, and the radius is $r = 0.5$ mm:

$$J = \sigma E = \left(5.8 \times 10^7\right)(0.01) = 5.8 \times 10^5 \text{ A/m}^2 \quad \text{[from (1.1.45.2)]} \tag{4.1}$$

$$I = JA = 5.8 \times 10^5 \cdot \pi \left(5 \times 10^{-4}\right)^2 \tag{4.2}$$

$$= 0.4555 \text{ A} \approx 0.46 \text{ A} \tag{4.3}$$

figs/Copper_wire.png

Fig. 4.1: Current in the wire against its diameter

Consider a non-negative function $f(x)$, continuous and bounded on $[2, 8]$. Let M and m be its maximum and minimum on the interval. Choose the combination(s) of α and β for which

$$\beta \leq \int_2^8 f(x)\,\mathrm{d}x \leq \alpha \tag{4.4}$$

is guaranteed to hold. (EC 2025)

(A) $\beta = 5m,\ \alpha = 7M$
(B) $\beta = 6m,\ \alpha = 5M$
(C) $\beta = 7m,\ \alpha = 6M$
(D) $\beta = 7m,\ \alpha = 5M$

Solution: The interval has length $8-2=6$, so

$$6m \le \int_2^8 f(x)\,\mathrm{d}x \le 6M \qquad \text{[from (1.1.49.1)]} \tag{4.1}$$

The inequality $\beta \le \int f \le \alpha$ is guaranteed when $\beta \le 6m$ and $\alpha \ge 6M$ (with $m, M \ge 0$). Option (A) has $5m \le 6m$ and $7M \ge 6M$; (B) has $\alpha = 5M < 6M$; (C) has $\beta = 7m > 6m$; (D) fails both.

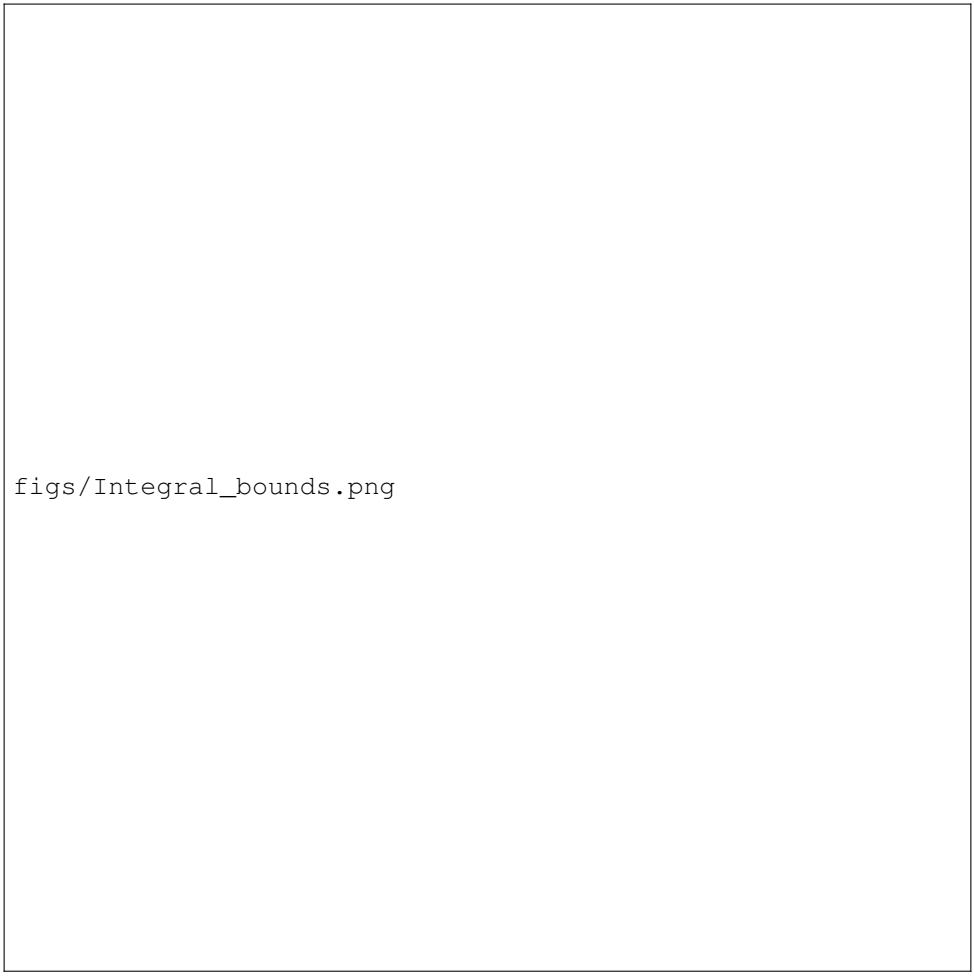


Fig. 4.1: The integral lies between $6m$ and $6M$

Which of the following statements about contour integrals (counter-clockwise) on the unit circle C in the complex plane is/are TRUE? (EC 2025)

(A) $\oint_C e^z \, dz = 0$

(B) $\oint_C z^n \, dz = 0$, where n is an even integer

(C) $\oint_C \cos z \, dz \neq 0$

(D) $\oint_C \sec z \, dz \neq 0$

Solution: The functions e^z and $\cos z$ are analytic everywhere, so both integrals vanish (1.1.52.1):

$$\oint_C e^z \, dz = 0 \qquad \text{[from (1.1.52.1)]} \tag{4.1}$$

$$\oint_C \cos z \, dz = 0 \qquad \text{[from (1.1.52.1)]} \tag{4.2}$$

(A) is true and (C) is false. For integer $n \neq -1$, in particular every even n,

$$\oint_C z^n \, dz = 0 \qquad \text{[from (1.1.53.1)]} \tag{4.3}$$

so (B) is true. The poles of $\sec z$ are at $\pm\pi/2, \pm 3\pi/2, \ldots$ (1.1.54.1), all outside $|z| = 1$, so

$$\oint_C \sec z \, dz = 0 \qquad \text{[from (1.1.52.1)]} \tag{4.4}$$

and (D) is false.

figs/Contour_integrals.png

Fig. 4.1: Unit circle and the poles of $\sec z$ and $1/z$

Consider a system with internal states $x_1(t), x_2(t), x_3(t)$ and input $u(t)$ governed by

$$\frac{d}{dt}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} u(t) \tag{4.5}$$

Which of the following statements is/are TRUE? (EC 2025)

(A) The signals x_1, x_2, x_3 are bounded for all bounded inputs.

(B) There exists a bounded input such that at least one of x_1, x_2, x_3 is unbounded.

(C) There exists a bounded input such that the signals x_1, x_2, x_3 are unbounded.

(D) The signals x_1, x_2, x_3 are unbounded for all bounded inputs.

Solution: Since $\mathbf{A}$ is diagonal, the states decouple:

$$\dot{x}_1 = 2x_1 + u \tag{4.1}$$

$$\dot{x}_2 = -2x_2 + u \tag{4.2}$$

$$\dot{x}_3 = u \tag{4.3}$$

For $u \equiv 1$ and zero initial state, x_1 (eigenvalue $+2$) and x_3 (eigenvalue 0, an integrator) grow without bound, while x_2 (eigenvalue -2) stays bounded for every bounded input:

$$x_1(t) = \frac{1}{2}\left(e^{2t} - 1\right) \to \infty \qquad \text{[from (1.1.31.3)]} \tag{4.4}$$

$$x_3(t) = t \to \infty \qquad \text{[from (1.1.31.4)]} \tag{4.5}$$

$$|x_2(t)| \le |x_2(0)| + \frac{U}{2} \qquad \text{[from (1.1.31.2)]} \tag{4.6}$$

So (B) is true. (C) and (D) need all three signals unbounded, which fails because of x_2; (A) is false.



Fig. 4.1: State responses to $u \equiv 1$ with zero initial state

The random variable X takes values in $\{-1, 0, 1\}$ with $\Pr(X = -1) = \Pr(X = 1) = \alpha$ and $\Pr(X = 0) = 1 - 2\alpha$, where $0 < \alpha < \frac{1}{2}$. Let $g(\alpha)$ be the entropy of X in bits. Which of the following is/are TRUE? (EC 2025)

(A) $g(0.4) > g(0.3)$
(B) $g(0.3) > g(0.4)$
(C) $g(0.3) > g(0.25)$
(D) $g(0.25) > g(0.3)$

Solution:

$$g(\alpha) = -2\alpha \log_2 \alpha - (1 - 2\alpha) \log_2 (1 - 2\alpha) \quad \text{[from (1.1.41.1)]} \tag{4.1}$$

$$g'(\alpha) = 2 \log_2 \frac{1 - 2\alpha}{\alpha} \tag{4.2}$$

$g' > 0$ for $\alpha < \frac{1}{3}$, so g increases up to $\frac{1}{3}$: $g(0.3) > g(0.25)$. The points 0.3 and 0.4 lie on opposite sides of the maximum, so evaluate directly:

$$g(0.25) = 0.5 \cdot 2 + 0.5 \cdot 1 = 1.5000 \tag{4.3}$$

$$g(0.3) = -0.6 \log_2 0.3 - 0.4 \log_2 0.4 = 1.5710 \tag{4.4}$$

$$g(0.4) = -0.8 \log_2 0.4 - 0.2 \log_2 0.2 = 1.5219 \tag{4.5}$$

Hence $g(0.3) > g(0.4)$ and $g(0.3) > g(0.25)$.

figs/Entropy_ternary.png

Fig. 4.1: Entropy $g(\alpha)$ with $\alpha = 0.25, 0.3, 0.4$ marked

Let $f(t)$ be periodic with fundamental period $T_0 > 0$, and $y(t) = f(\alpha t)$ with $\alpha > 1$. Their

Fourier series are

$$f(t) = \sum_{k=-\infty}^{\infty} c_k \, e^{\mathrm{j}\frac{2\pi}{T_0}kt}, \qquad y(t) = \sum_{k=-\infty}^{\infty} d_k \, e^{\mathrm{j}\frac{2\pi\alpha}{T_0}kt} \tag{4.6}$$

Which of the following is/are TRUE? (EC 2025)

(A) $c_k = d_k$ for all k
(B) $y(t)$ is periodic with fundamental period αT_0
(C) $c_k = d_k/\alpha$ for all k
(D) $y(t)$ is periodic with fundamental period T_0/α

Solution: Since $y\left(t + T_0/\alpha\right) = f\left(\alpha t + T_0\right) = y(t)$, the period of $y(t) = f(\alpha t)$ is T_0/α (1.1.22.3). Replacing t by αt in the series of f:

$$y(t) = f(\alpha t) = \sum_{k} c_k \, e^{\mathrm{j}2\pi k\alpha t/T_0} \qquad \text{[from (1.1.22.2)]} \tag{4.1}$$

Comparing with the series of $y(t)$ gives $d_k = c_k$ for all k.

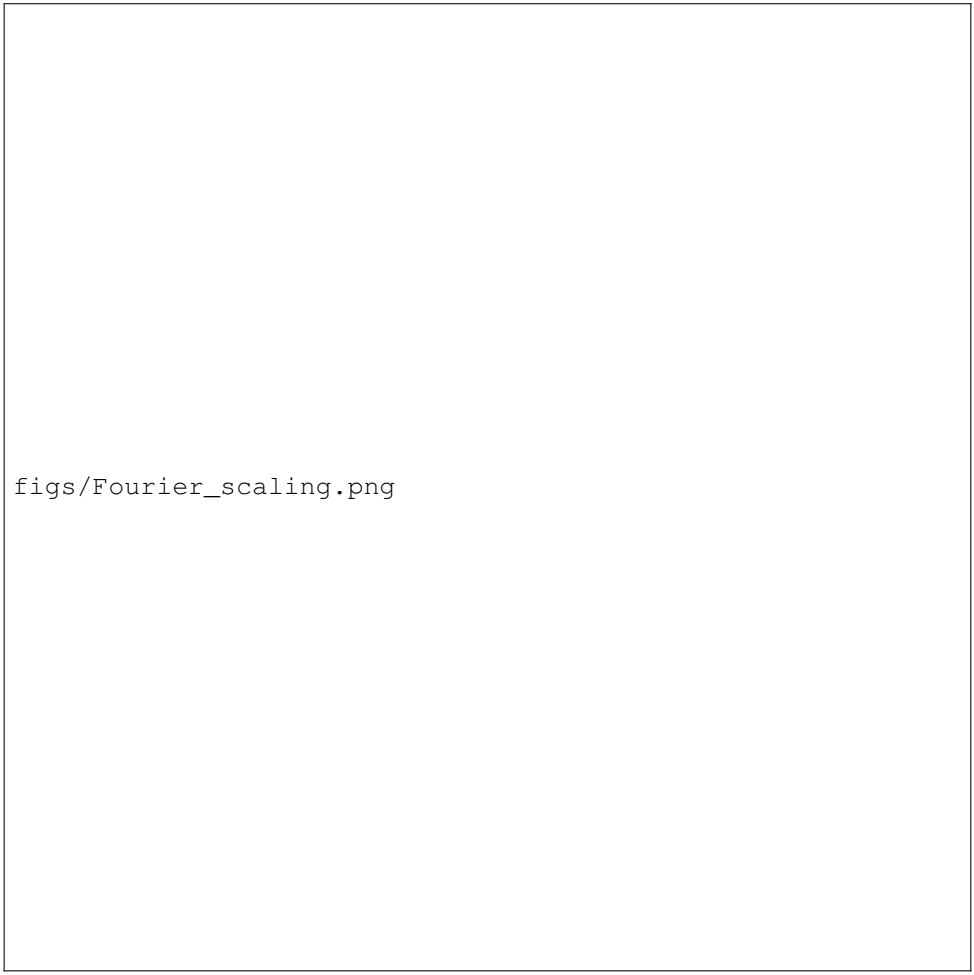


Fig. 4.1: $f(t)$ and $f(3t)$ have identical Fourier coefficients

Consider the system represented by the block diagram in Fig. 4.1. Which of the signal flow graphs in Fig. 4.2 represent(s) this system? Choose the correct option(s). (EC 2025)

(A) graph (A)
(B) graph (B)
(C) graph (C)
(D) graph (D)

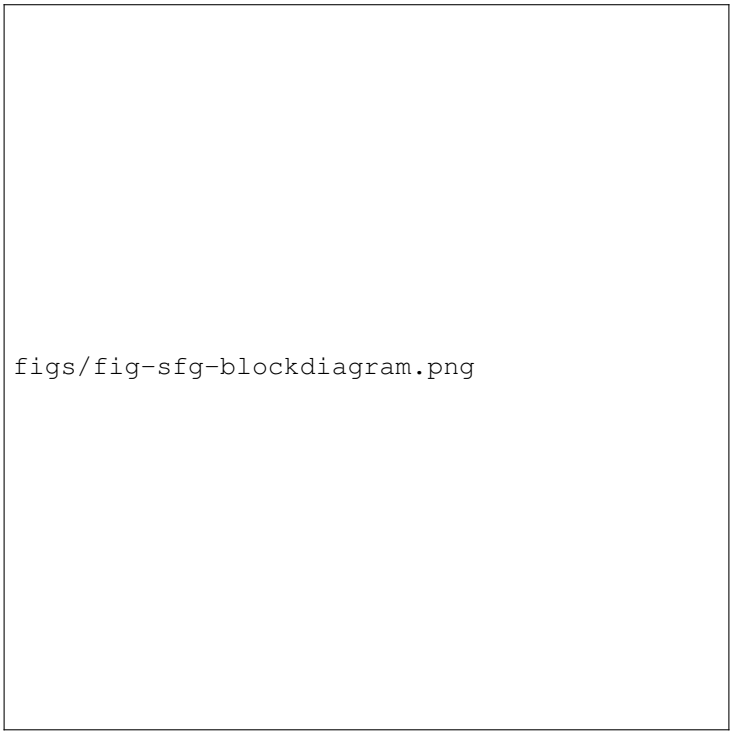


Fig. 4.1: Block diagram of the system

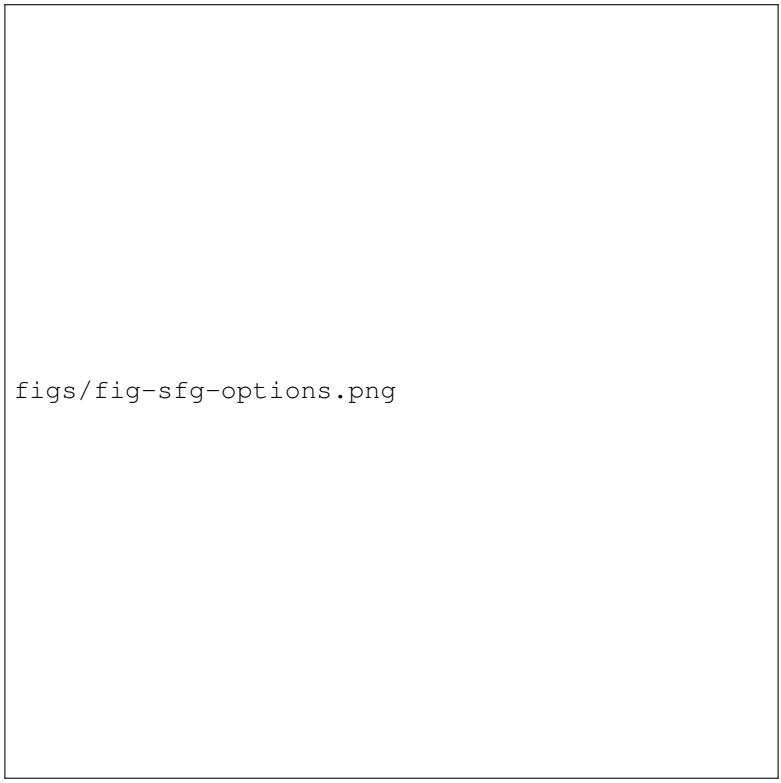


Fig. 4.2: Candidate signal flow graphs (A)–(D)

Solution: From the block diagram, the first junction gives $e = R - G_1 e$ and the output is $Y = (G_1 + G_2)\, e$:

$$e = \frac{R}{1 + G_1} \tag{4.1}$$

$$\frac{Y}{R} = \frac{G_1 + G_2}{1 + G_1} \tag{4.2}$$

Applying Mason's formula (1.1.30.1) to each graph (forward paths G_1, G_2 in all four):

graph	loop gains	$T = Y/R$
(A)	$-G_1,\ -G_2$	$\dfrac{G_1 + G_2}{1 + G_1 + G_2}$
(B)	$-G_1$	$\dfrac{G_1 + G_2}{1 + G_1}$
(C)	$+G_1,\ +G_2$	$\dfrac{G_1 + G_2}{1 - G_1 - G_2}$
(D)	$-G_1,\ -G_2$	$\dfrac{G_1 + G_2}{1 + G_1 + G_2}$

Only graph (B), whose single loop is G_1 closed by -1, equals $\dfrac{G_1+G_2}{1+G_1}$.

Two fair dice (faces $1,\ldots,6$) are rolled. Let the random variable X be the sum of the outcomes. The expectation of X is _____ (rounded off to two decimal places). (EC 2025)

Solution: Let $X = X_1 + X_2$, where X_i is the outcome of die i:

$$E[X_i] = \frac{1+2+3+4+5+6}{6} = 3.5 \tag{4.3}$$

$$E[X] = E[X_1] + E[X_2] = 7.00 \qquad \text{[from (1.1.6.2)]} \tag{4.4}$$

figs/Dice_sum.png

Fig. 4.3: Distribution of the sum of two dice

Consider the vectors

$$\mathbf{a} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 0 \\ 3\sqrt{2} \end{pmatrix} \tag{4.5}$$

For a real scalar x, the value of $\min_x \|x\mathbf{a} - \mathbf{b}\|_2$ is _____ (rounded off to two decimal places), where $\|\mathbf{y}\|_2 = \sqrt{y_1^2 + y_2^2}$. (EC 2025)

Solution:

$$x^* = \frac{\mathbf{a}^\top \mathbf{b}}{\mathbf{a}^\top \mathbf{a}} = \frac{3\sqrt{2}}{2} \qquad \text{[from (1.1.51.1)]} \tag{4.6}$$

$$x^*\mathbf{a} - \mathbf{b} = \begin{pmatrix} \frac{3\sqrt{2}}{2} \\ -\frac{3\sqrt{2}}{2} \end{pmatrix} \tag{4.7}$$

$$\|x^*\mathbf{a} - \mathbf{b}\| = \sqrt{2 \cdot \frac{18}{4}} = 3.00 \tag{4.8}$$

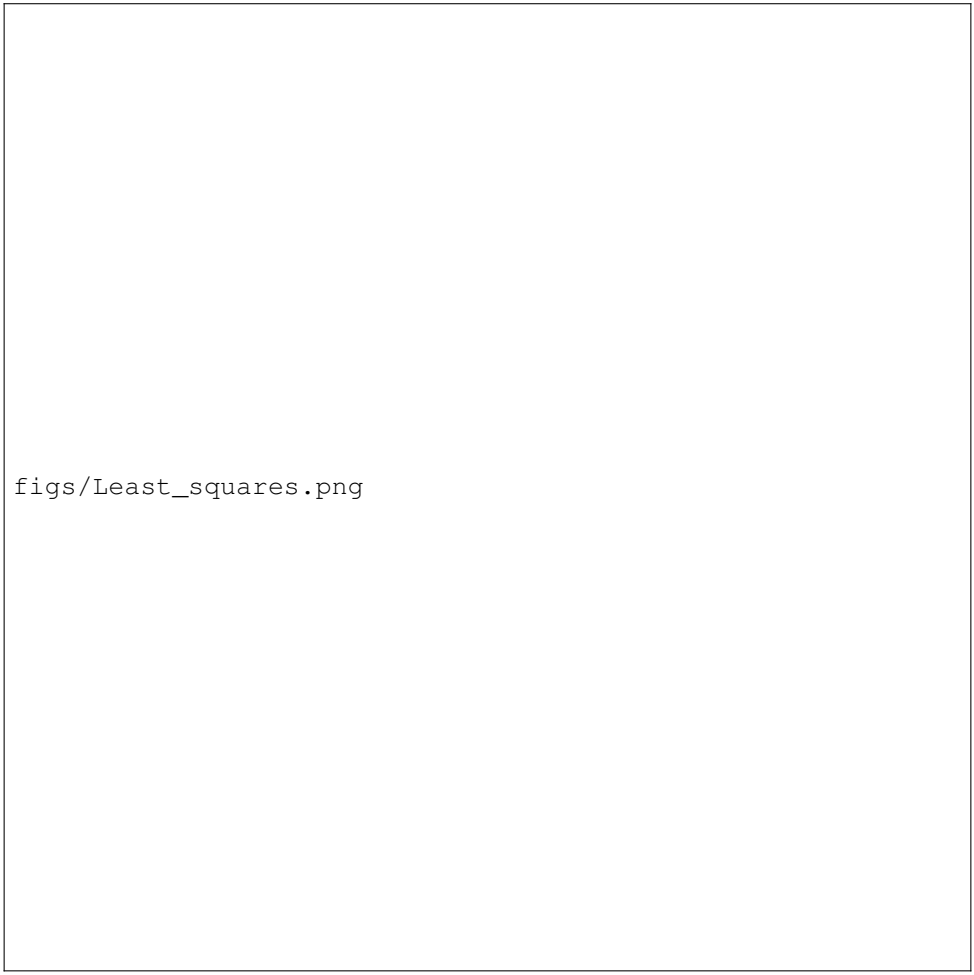


Fig. 4.4: Projection of $\mathbf{b}$ on the line through $\mathbf{a}$

X and Y are Bernoulli random variables on $\{0,1\}$ with joint pmf

$$\Pr(0,0)=0.06, \quad \Pr(0,1)=0.14, \quad \Pr(1,0)=0.24, \quad \Pr(1,1)=0.56 \tag{4.9}$$

The mutual information $I(X;Y)$ is _____ (rounded off to two decimal places). (EC 2025)

Solution: The marginals follow by summing the joint PMF (1.1.7.1):

$$\Pr(X=0)=0.06+0.14=0.20 \tag{4.10}$$
$$\Pr(X=1)=0.24+0.56=0.80 \tag{4.11}$$
$$\Pr(Y=0)=0.06+0.24=0.30 \tag{4.12}$$
$$\Pr(Y=1)=0.14+0.56=0.70 \tag{4.13}$$

Every joint probability equals the product of the marginals, so X and Y are independent (1.1.8.1):

$$\Pr(X=0)\Pr(Y=0) = 0.2 \cdot 0.3 = 0.06 \tag{4.14}$$

$$\Pr(X=1)\Pr(Y=1) = 0.8 \cdot 0.7 = 0.56 \tag{4.15}$$

$$I(X;Y) = 0 \qquad \text{[from (1.1.42.2)]} \tag{4.16}$$

figs/Mutual_information.png

Fig. 4.5: Joint PMF against the product of the marginals

The diode in the circuit of Fig. 4.6 is ideal. The input voltage (in Volts) is $V_I = 10\sin(100\pi t)$, with t in seconds. The time duration (in ms, two decimal places) for which the diode is forward biased during one period of the input is _____. (EC 2025)

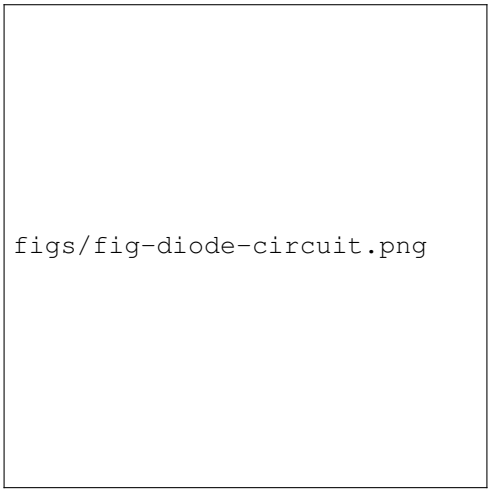


Fig. 4.6: Ideal diode circuit

Solution: From the figure the battery's negative plate faces the diode, so the cathode is at -5 V and the diode conducts when $V_I > -5$ V. With $\theta = 100\pi t$:

$$10\sin\theta > -5 \iff \sin\theta > -\frac{1}{2} \tag{4.17}$$

$$\Delta\theta = \frac{7\pi}{6} - \left(-\frac{\pi}{6}\right) = \frac{4\pi}{3} \tag{4.18}$$

$$\Delta t = \frac{\Delta\theta}{\omega} = \frac{4\pi/3}{100\pi} = \frac{1}{75}\ \text{s} = 13.33\ \text{ms} \qquad \text{[from (1.1.47.1)]} \tag{4.19}$$

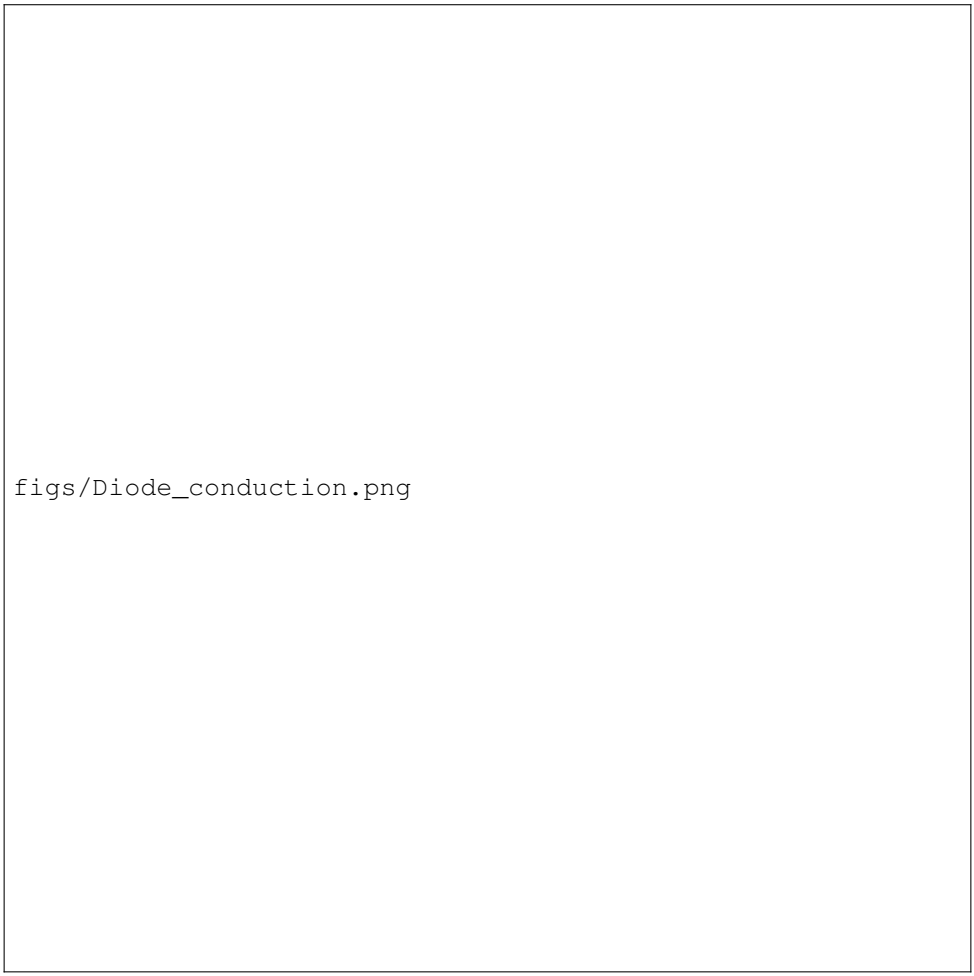


Fig. 4.7: Conduction interval of the diode

In the circuit of Fig. 4.8, the AND gate has a propagation delay of 1 ns. The edge-triggered flip-flops have set-up time 2 ns, hold time 0 ns and clock-to-Q delay 2 ns. The maximum clock frequency (in MHz, nearest integer) with no setup violation is _____. (EC 2025)

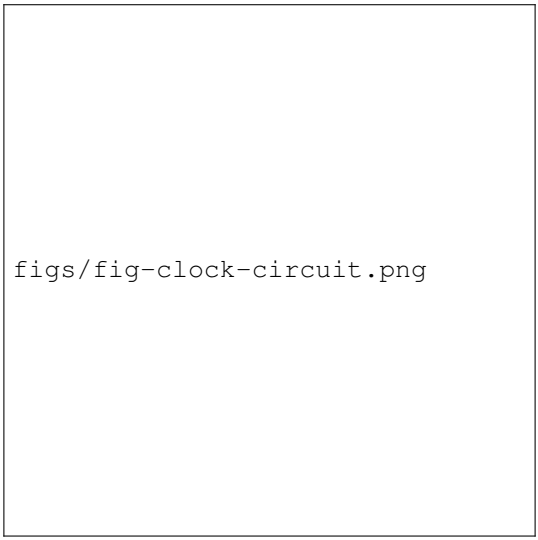


Fig. 4.8: Flip-flops with an AND gate in the data path

Solution: From the figure the longest path is $\text{FF}_2 \rightarrow \text{AND} \rightarrow \text{FF}_1$:

$$T_{\text{clk,min}} = t_{cq} + t_{AND} + t_{setup} = 2 + 1 + 2 = 5 \text{ ns} \qquad \text{[from (1.1.35.3)]} \tag{4.20}$$

$$f_{\max} = \frac{1}{5 \text{ ns}} = 200 \text{ MHz} \tag{4.21}$$

Two resistors are connected in a circuit loop of area $5\ \text{m}^2$, as in Fig. 4.9. The loop lies in the x–y plane. A time-varying magnetic flux density $B(t) = 0.5t$ (in Tesla) is applied along the positive z-axis. The magnitude of the current I (in Amperes, two decimal places) in the loop is _____. (EC 2025)



Fig. 4.9: Circuit loop with two resistors

Solution: The two 2 Ω resistors in the loop are in series, so $R_{\text{total}} = 4\ \Omega$:

$$|\mathcal{E}| = A\left|\frac{dB}{dt}\right| = 5 \times 0.5 = 2.5\ \text{V} \qquad \text{[from (1.1.44.2)]} \tag{4.22}$$

$$I = \frac{|\mathcal{E}|}{R_{\text{total}}} = \frac{2.5}{4} = 0.625\ \text{A} \approx 0.63\ \text{A} \tag{4.23}$$

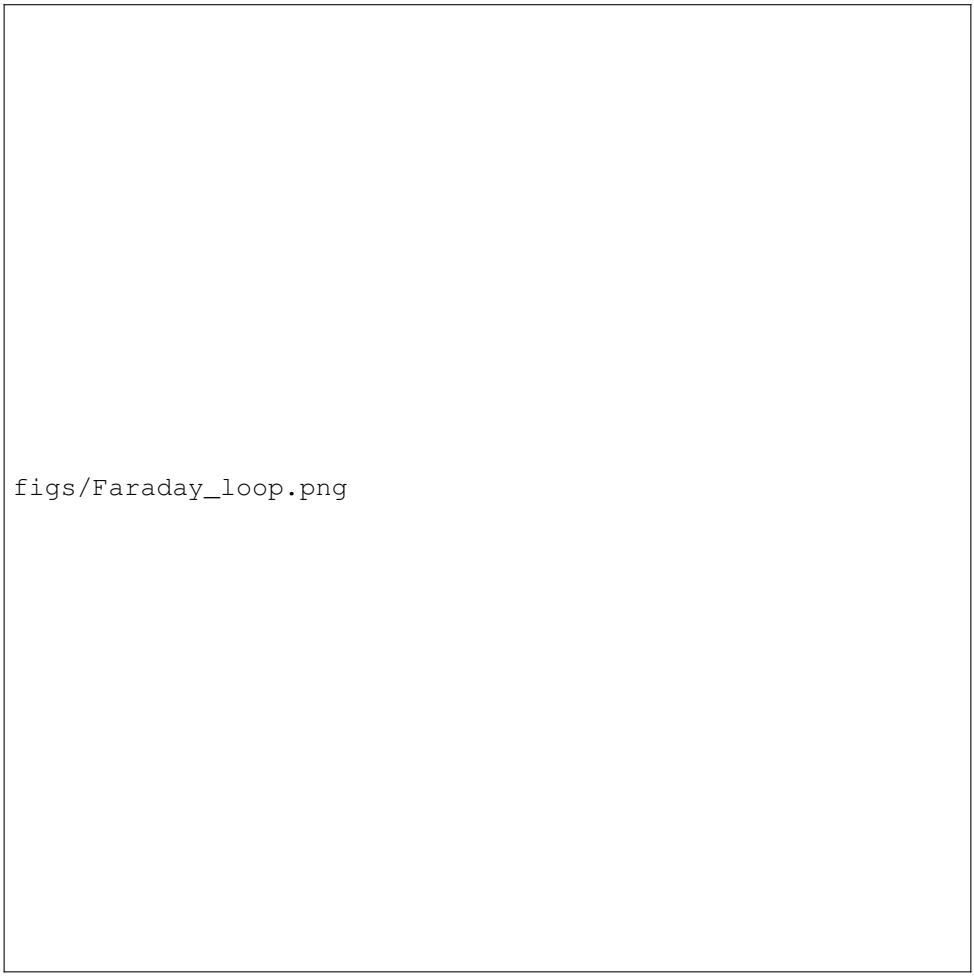


Fig. 4.10: Flux, emf and current in the loop